

High-performance Sub-1GHz RF Transceiver

Features

- Frequency range: 113 – 960 MHz
- Modulation/demodulation: OOK, 2 (G)FSK, 4 (G)FSK, G(MSK)
- Data rate: 0.1 - 1000 kbps
- Sensitivity: 2 FSK, -122 dBm DR=2.4 kbps, 433.92 MHz
4 FSK, -88 dBm DR=1 Mbps, 433.92 MHz
OOK, -94 dBm DR= 300 kbps, 433.92 MHz
- Adjacent channel rejection : 62 dBc, BW = 4.8 kHz,
Channel space = 12.5 kHz
- Blocking: 76 dBc , ± 1 MHz offset, BW = 4.8 kHz
- Operating voltage range: 1.8 - 3.6 V
- Tx current: 30 mA @ 13 dBm, 433.92 MHz, FSK
82 mA @ 20 dBm, 433.92 MHz, FSK
- Rx current: 9.6 mA (DCDC) @433.92 MHz, FSK
- No extra RF switching components required in single antenna mode
- Multiple super-low power (SLP) Rx modes available
- Sleep current
 - ◆ 400 nA, Duty Cycle = OFF
 - ◆ 800 nA, Duty Cycle = ON
- Special features:
 - ◆ Quick and stable automatic frequency control (AFC)
 - ◆ 3 varied clock data recovery systems (CDR)
 - ◆ Quick and accurate valid signal detection (PJD, RSSI)
 - ◆ Super-low power (SLP) and Duty Cycle receiving
 - ◆ Fast Tx or Rx frequency hopping
 - ◆ Carrier sense multiple access (CSMA)
 - ◆ Automatic ACK and re-sending
 - ◆ Antenna diversity
- 4-wire SPI interface
- Supporting both direct and packet modes, with configurable packet handler and 128-Byte FIFO
- NRZ format, Manchester, whitened data coding, FEC

Application

- Automatic meter reading
- Home security and building automation
- ISM-band data communication
- Industrial monitoring and control
- Remote control and security system
- Remote key entry
- Wireless sensor node
- Tag reader and writer

Ordering Information

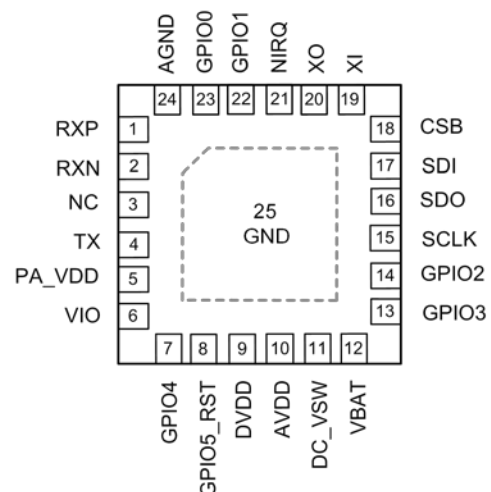
Product Model	Frequency	Package	Minimum Order
CMT2312A-EQR	113 - 960 MHz	QFN24	3,000 pcs
Refer to Table 22 for more ordering information.			



QF24 (4X4)

Description

The CMT2312A is an ultra-low power, high-performance, OOK / 2(G)FSK / 4(G)FSK / G(MSK) based RF transceiver, applicable to various applications within the 113 - 960 MHz frequency band. The product is part of the CMOSTEK NextGenRF™ product family which covers a complete product line consisting of transmitters, receivers and transceivers. The high-density integration of CMT2312A simplifies the required BOM in system design. With Tx power reaching +20 dBm and sensitivity reaching -122 dBm, it can achieve optimized performance of application RF links. Through providing multiple data packet formats and code methods, this product ensures the flexible supporting of various applications. Besides, the CMT2312A provides functions such as 128-byte Tx/Rx FIFO, multiple GPIO and interrupt configurations, Duty-Cycle mode, LBT (listen before talk), high-precision RSSI, LBD, power on reset, low-frequency clock output, quick frequency hopping, squelch, etc., which allows more flexible application design and gains more product differentiation capability.



CMT2312A Top View

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1 Electrical Specifications

The measurement conditions are: $V_{DD} = 3.3\text{ V}$, $T_{OP} = 25\text{ }^{\circ}\text{C}$, $F_{RF} = 433.92\text{ MHz}$, sensitivity is measured by receiving a PN9 sequence, matching to $50\text{ }\Omega$ impedance and 0.1% BER if nothing else stated. All measurement results are obtained using the evaluation board CMT2312A-EM if nothing else stated.

1.1 Recommended Operating Conditions

Table 1. Recommended Operating Conditions

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Operating supply voltage	V_{DD}		1.8		3.6	V
Operating temperature	T_{OP}		-40		85	$^{\circ}\text{C}$
Supply voltage slope			1			mV/us

1.2 Absolute Maximum Ratings

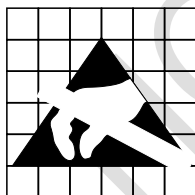
Table 2. Absolute Maximum Ratings^[1]

Parameter	Symbol	Condition	Min.	Typ.	Max.
Supply voltage	V_{DD}		-0.3	3.6	V
Interface voltage	V_{IN}		-0.3	3.6	V
Junction temperature	T_J		-40	125	$^{\circ}\text{C}$
Storage temperature	T_{STG}		-50	150	$^{\circ}\text{C}$
Soldering temperature	T_{SDR}	Last for at least 30 seconds Human body model (HBM)		255	$^{\circ}\text{C}$
ESD rating ^[2]		Human body model (HBM)	-2	2	kV
Latch-up current		@ $85\text{ }^{\circ}\text{C}$	-100	100	mA

Notes:

[1]. Exceeding the Absolute Maximum Ratings may cause permanent damage to the equipment. This value is a pressure rating and does not imply that the function of the equipment is affected under this pressure condition, but if it is exposed to absolute maximum ratings for extended periods of time, it may affect equipment reliability.

[2]. The CMT2312A is a high performance RF integrated circuit. The operation and assembly of this chip should only be performed on a workbench with good ESD protection.



Caution! ESD sensitive device. Precaution should be used when handling the device in order to prevent performance degradation or loss of functionality.

1.3 RF Power Consumption

Table 3. RF Power Consumption

Parameter	Symbol	Condition	Typ. (Disable DCDC)	Typ. (Enable DCDC)	Unit
Sleep current ^[1]	I_{SLEEP}	In sleep mode with sleep timer disabled	400		nA
		In sleep mode with sleep timer enabled	800		nA

Parameter	Symbol	Condition		Typ. (Disable DCDC)	Typ. (Enable DCDC)	Unit
Ready current ^[1]	I _{Ready}			2.1	1.9	mA
RFS current ^[1]	I _{RFS}		315 MHz	7.5	5.2	mA
			433 MHz	7.8	5.6	mA
			868 MHz	8.4	5.9	mA
			915 MHz	8.5	5.9	mA
TFS current ^[1]	I _{TFS}		315 MHz	7.5	5.2	mA
			433 MHz	7.8	5.6	mA
			868 MHz	8.4	5.9	mA
			915 MHz	8.5	5.9	mA
RX current ^[1]	I _{Rx}	DR = 10 kbps Dev =10 kHz	315 MHz	13.5	8.8	mA
			433 MHz	13.6	9.4	mA
			868 MHz	14.3	9.9	mA
			915 MHz	14.3	9.9	mA
TX current ^[1]	I _{Tx}	20 dBm ^[2]	315 MHz	74	/	mA
			433 MHz	82	81	mA
			868 MHz	88	87	mA
			915 MHz	88	87	mA
		13 dBm ^[3]	315 MHz	26.7	/	mA
			433 MHz	30	29	mA
			868 MHz	33	32	mA
			915 MHz	34	33	mA
		10 dBm ^[3]	315 MHz	21	15	mA
			433 MHz	25	24	mA
			868 MHz	27	26	mA
			915 MHz	27	26	mA
		-10 dBm ^[3]	315 MHz	10.3	7	mA
			433 MHz	11	10	mA
			868 MHz	12	11	mA
			915 MHz	12	11	mA

Notes:

[1]. 2FSK, DR = 10 kbps, F_{DEV} = 5 kHz, Vbat = 3.3 V.

[2]. Apply 20 dBm matching network.

[3]. Apply 13 dBm matching network.

1.4 Receiver

Table 4. Receiver Specifications

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Data rate	DR	OOK	0.1		300	kbps
		2 (G)FSK	0.1		500	kbps
		4(GFSK	0.1		1000	kbps
Deviation (RX)	F _{DEV}	(G)FSK, 4(G)FSK ^[1]	0.5		350	kHz
Sensitivity @ 433 MHz (direct tiematching network)	S ₄₃₃	FSK ^[2]	DR = 2.4 kbps, F _{DEV} = 1.2 kHz, BW= 4.8kHz		-122	dBm
			DR = 10 kbps, F _{DEV} = 5 kHz		-114	dBm
			DR = 20 kbps, F _{DEV} = 10 kHz		-112	dBm
			DR = 50 kbps, F _{DEV} = 25 kHz		-109	dBm
			DR =100 kbps, F _{DEV} = 50 kHz		-106	dBm
			DR =200 kbps, F _{DEV} = 100 kHz		-104	dBm
			DR =500 kbps, F _{DEV} = 250 kHz		-98	dBm

Parameter	Symbol	Condition		Min.	Typ.	Max.	Unit		
		OOK ^[2]	5 kbps		-110		dBm		
			50 kbps		-101		dBm		
			100 kbps		-97		dBm		
			200 kbps		-95		dBm		
			300 kbps		-94		dBm		
		4FSK ^[2]	DR = 10 kbps, F _{DEV} ^[3] = 10kHz		-109		dBm		
			DR = 100 kbps, F _{DEV} ^[3] =100kHz		-99		dBm		
			DR = 1 Mbps, F _{DEV} ^[3] = 250 kHz		-88		dBm		
		Sensitivity @ 868 MHz (direct tie matching network)	S ₈₆₈	FSK ^[2]	DR = 2.4 kbps, F _{DEV} = 1.2 kHz, BW=4.8kHz		-120		dBm
					DR = 10 kbps, F _{DEV} = 5 kHz		-111		dBm
DR = 20 kbps, F _{DEV} = 10 kHz					-110		dBm		
DR = 50 kbps, F _{DEV} = 25 kHz					-107		dBm		
DR =100 kbps, F _{DEV} = 50 kHz					-104		dBm		
DR =200 kbps, F _{DEV} = 100 kHz					-102		dBm		
DR =500 kbps, F _{DEV} = 250 kHz					-96		dBm		
OOK ^[2]	5 kbps				-106		dBm		
	50 kbps				-98		dBm		
	100 kbps				-94		dBm		
	200 kbps				-93		dBm		
	300 kbps				-92		dBm		
4FSK ^[2]	DR = 10 kbps, F _{DEV} ^[3] = 10kHz				-106		dBm		
	DR = 100 kbps, F _{DEV} ^[3] = 100kHz				-96		dBm		
	DR = 1 Mbps, F _{DEV} ^[3] = 250 kHz				-85		dBm		
Notes:									
[1]. BT = 0.5 by default for Gaussian modulation.									
[2]. In case of unspecified BW value, a crystal of 10 ppm is used and the BW value is automatically calculated by RFPDK.									
[3]. For 4 FSK, F _{DEV} represents the frequency difference between the frequency points at the far ends (left and right) and the centered frequency point.									
Receiver channel bandwidth	BW	Receiver channel bandwidth		1.3		1168	kHz		
Saturation input signal level	P _{LVL}					20	dBm		
RSSI range	RSSI	By a step of 1 dB		-127		20	dBm		
Co-channel rejection @ 433 MHz, 868 MHz	CCR	DR = 2.4 kbps; F _{DEV} = 1.2 kHz; BW=4.8 kHz CW interference, BER<0.1%			-7		dB		
Adjacent channel rejection @ 433 MHz	ACR-I ₄₃₃	DR = 2.4 kbps; F _{DEV} = 1.2 kHz; BW= 4.8 kHz, Channel Space = 12.5 kHz, CW interference, BER<0.1%			62		dB		
Adjacent channel rejection @ 868 MHz	ACR-I ₈₆₈	DR = 2.4 kbps; F _{DEV} = 1.2 kHz; BW= 4.8 kHz, Channel Space = 12.5 kHz, CW interference, BER<0.1%			56		dB		
Blocking @ 433 MHz	BI ₄₃₃	DR = 2.4 kbps; F _{DEV} = 1.2 kHz; BW=4.8 kHz, CW interference, BER<0.1%	±1 MHz offset		76		dB		
			±2 MHz offset		80		dB		
			±10 MHz offset		84		dB		

Parameter	Symbol	Condition		Min.	Typ.	Max.	Unit
Blocking @ 868 MHz	BI ₈₆₈	DR = 2.4 kbps; F _{DEV} = 1.2 kHz; BW=4.8 kHz, CW interference, BER<0.1%	±1 MHz offset		66		dB
			±2 MHz offset		76		dB
			±10 MHz offset		83		dB
Image Rejection @ 433 MHz	IMR ₄₃₃	DR = 2.4 kbps; F _{DEV} = 1.2 kHz; BW=4.8 kHz CW interference, BER<0.1%	Before calibration		30		dB
			After calibration		56		dB
Image Rejection @ 868 MHz	IMR ₈₆₈	DR = 2.4 kbps; F _{DEV} = 1.2 kHz; BW=4.8 kHz CW interference, BER<0.1%	Before calibration		26		dB
			After calibration		51		dB
Input 3rd order intercept point @ 433 MHz	IIP3 ₄₃₃	DR = 2.4 kbps; F _{DEV} = 1.2 kHz; two-tone test with 10 MHz and 20 MHz deviations.			-13		dBm
Input 3rd order intercept point @ 868 MHz	IIP3 ₈₆₈	DR = 2.4 kbps; F _{DEV} = 1.2 kHz; two-tone test with 10 MHz and 20 MHz deviations.			-12		dBm
Receiver input impedance	Z _{in}	RXP and RXN Differential input impedance	433MHz	150Ω// 0.8pF			
			868MHz	134Ω// 1.0pF			

1.5 Transmitter

Table 5. Transmitter Specifications

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Output power	P _{OUT}	Specific peripheral components are required according to different frequency bands.	-10		+20	dBm
Output power step	P _{STEP}			1		dB
GFSK Gaussian filter coefficient	BT		0.3	0.5	1.0	-
Output power change in different temperature	P _{OUT-TOP}	Temperature range: -40 to +85 °C		1		dB
Spurious emissions		P _{OUT} = +13 dBm, 433 MHz, F _{RF} <1 GHz			-54	dBm
		1 GHz to 12.75 GHz, including Harmonic			-36	dBm
Harmonic output ^[1] for F _{RF} = 315 MHz	H2 ₃₁₅	2 nd harmonic, +20 dBm P _{OUT}		-57		dBm
	H3 ₃₁₅	3 rd harmonic, +20 dBm P _{OUT}		-75		dBm
Harmonic output ^[1] for F _{RF} = 433 MHz	H2 ₄₃₃	2 nd harmonic, +20 dBm P _{OUT}		-56		dBm
	H3 ₄₃₃	3 rd harmonic, +20 dBm P _{OUT}		-71		dBm
Harmonic output ^[1] for F _{RF} = 868 MHz	H2 ₈₆₈	2 nd harmonic, +20 dBm P _{OUT}		-47		dBm
	H3 ₈₆₈	3 rd harmonic, +20 dBm P _{OUT}		-72		dBm
Harmonic output ^[1] for F _{RF} = 915 MHz	H2 ₉₁₅	2 nd harmonic, +20 dBm P _{OUT}		-47		dBm
	H3 ₉₁₅	3 rd harmonic, +20 dBm P _{OUT}		-73		dBm
Harmonic output ^[1] for F _{RF} = 315 MHz	H2 ₃₁₅	2 nd harmonic, +13 dBm P _{OUT}		-51		dBm
	H3 ₃₁₅	3 rd harmonic, +13 dBm P _{OUT}		-72		dBm
Harmonic output ^[1] for F _{RF} = 433 MHz	H2 ₄₃₃	2 nd harmonic, +13 dBm P _{OUT}		-44		dBm
	H3 ₄₃₃	3 rd harmonic, +13 dBm P _{OUT}		-58		dBm

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Harmonic output ^[1] for $F_{RF} = 868 \text{ MHz}$	H2 ₈₆₈	2 nd harmonic, +13 dBm P _{OUT}		-50		dBm
	H3 ₈₆₈	3 rd harmonic, +13 dBm P _{OUT}		-71		dBm
Harmonic output ^[1] for $F_{RF} = 915 \text{ MHz}$	H2 ₉₁₅	2 nd harmonic, +13 dBm P _{OUT}		-54		dBm
	H3 ₉₁₅	3 rd harmonic, +13 dBm P _{OUT}		-73		dBm
Notes: [1]. The harmonic level mainly depends on the quality of matching network. The parameters above are measured based on CMT2312A-EM.						

1.6 RF Frequency Synthesizer

Table 6. RF Frequency Synthesizer

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Frequency range	F _{RF}	Require different matching networks.	675		960	MHz
			338		640	MHz
			113		320	MHz
Frequency deviation range	F _{DEV_RNG} [1]	675 ~ 960 MHz		600		kHz
		450 ~ 640 MHz		400		kHz
		338 ~ 450 MHz		300		kHz
		225 ~ 320 MHz		200		kHz
		169 ~ 225 MHz		150		kHz
		135 ~ 169 MHz		120		kHz
		113 ~ 135 MHz		100		kHz
Frequency resolution	F _{RES}			60		Hz
Frequency tuning time	t _{TUNE}			60		us
Phase noise @ 433 MHz	PN ₄₃₃	10 kHz Frequency Offset		-101		dBc/Hz
		100 kHz Frequency Offset		-114		dBc/Hz
		1MHz Frequency Offset		-129		dBc/Hz
		10 MHz Frequency Offset		-134		dBc/Hz
Phase noise @ 868 MHz	PN ₈₆₈	10 kHz Frequency Offset		-100		dBc/Hz
		100 kHzFrequency Offset		-109		dBc/Hz
		1MHz Frequency Offset		-126		dBc/Hz
		10 MHz Frequency Offset		-129		dBc/Hz
Notes:						
[1]. For 2FSK and 4FSK, F _{DEV} represents the frequency difference between the frequency points at the far ends (left and right) and the centered frequency point.						

1.7 Settling Time

Table 7. Settling Time

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Settling time	T _{SLP-RX}	From Sleep to RX		660		us
	T _{SLP-TX}	From Sleep to TX		660		us

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
	T _{STB-RX}	From Standby to RX		160		us
	T _{STB-TX}	From Standby to TX		160		us
	T _{RFS-RX}	From RFS to RX		16		us
	T _{TFS-RX}	From TFS to TX		16		us
	T _{TX-RX}	From TX to RX (Ramp down requires 2T _{symbol} time)		2T _{symbol} +168		us
	T _{RX-TX}	From RX to TX		220		us
Notes:						
[1]. T _{SLP-RX} and T _{SLP-TX} mainly depend on crystal oscillating, which is largely related to crystal itself.						

1.8 Crystal Oscillator

Table 8. Crystal Specification

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Crystal frequency ^[1]	F _{XTAL}			32		MHz
Crystal frequency precision ^[2]	ppm _{XTAL}		0	20	100	ppm
Load resistance	C _{LOAD_XTAL}			12		pF
Crystal equivalent resistance	R _{mXTAL}			60		Ω
Crystal startup time ^[3]	t _{XTAL}			800		us
Notes:						
[1]. The CMT2312A can utilize external reference clock to directly drive XIN pin through the coupling capacitor. The peak-to-peak value of external clock signal is required between 0.3 and 0.7 V.						
[2]. It involves:(1) initial tolerance, (2) crystal loading, (3) aging, and (4) temperature changing. The acceptable crystal frequency tolerance is subject to the bandwidth of the receiver and the RF tolerance between the receiver and its paired transmitter.						
[3]. This parameter is to a large degree crystal dependent.						

1.9 Low-frequency Oscillator

Table 9. Low-frequency Oscillator Specifications

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Calibration frequency ^[1]	F _{LPOSC}			32		kHz
Frequency accuracy		After calibration		±1		%
Temperature coefficient ^[2]				-0.02		%/°C
Supply voltage coefficient ^[3]				+0.5		%/V
Initial calibration time	t _{LPOSC-CAL}			4		ms
Notes:						
[1]. The low-frequency oscillator is automatically calibrated to the crystal oscillator frequency at the PUP stage.						
[2]. After calibration, the frequency will drift with temperature.						
[3]. After calibration, the frequency will drift with the change of the supply voltage.						

1.10 Low Battery Detection

Table10. Low Battery Detection Specifications

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Detection accuracy	LBD _{RES}			50		mV

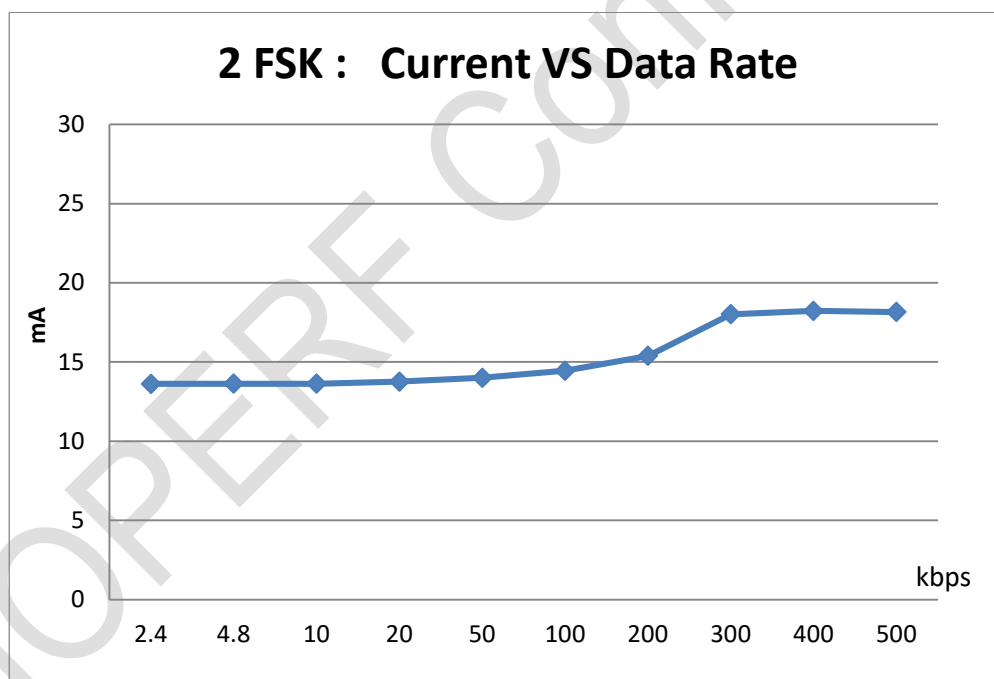
1.11 Digital Interface

Table 11. Digital Interface Specifications

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Digital input high level	V _{IH}		V _{DD} -0.4			V _{DD}
Digital input low level	V _{IL}				0.2	V _{DD}
Digital output high level	V _{OH}	@I _{OH} = -0.5 mA	V _{DD} -0.4			V
Digital output low level	V _{OL}	@I _{OL} = 0.5 mA			0.4	V
SCLK frequency	F _{SCL}				10	MHz
SCLK high time	T _{CH}		50			ns
SCLK low time	T _{CL}		50			ns
SCLK rising edge time	T _{CR}			10		ns
SCLK falling edge time	T _{CF}			10		ns

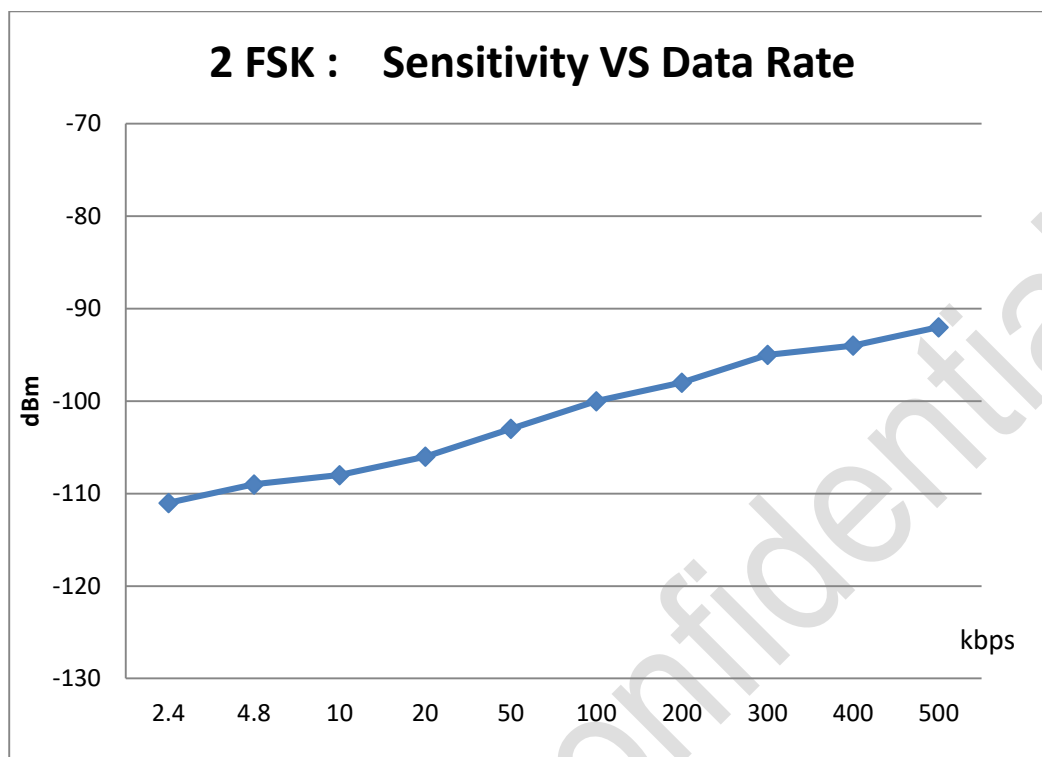
1.12 Typical Parameter Chart

1.12.1 Rx Current Vs. Data rate



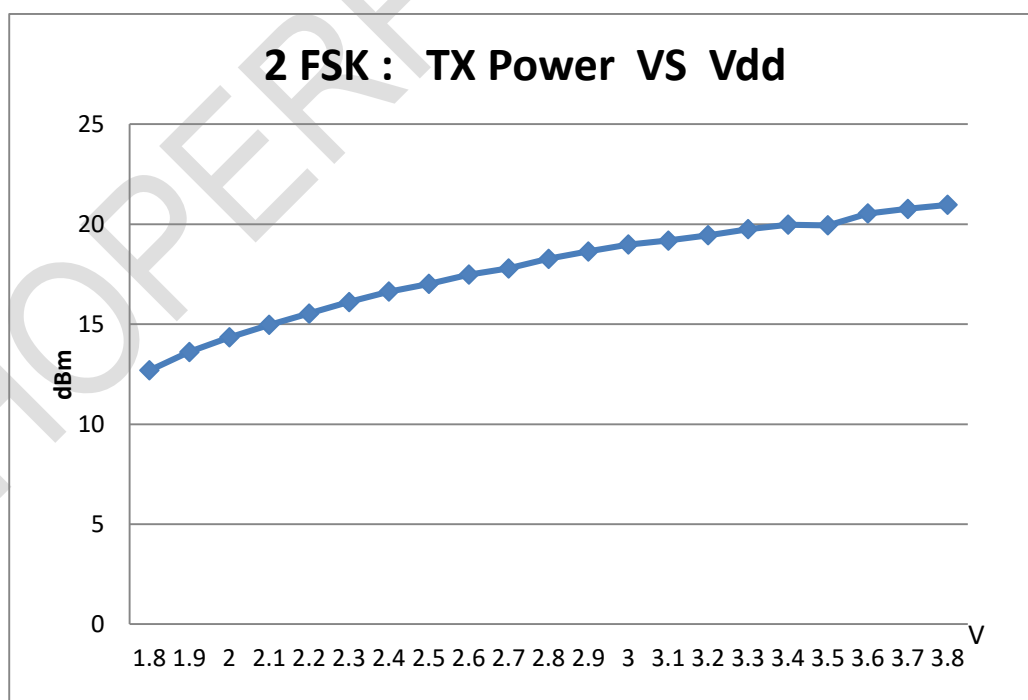
Test condition: Freq = 434 MHz, ppm = 10.

1.12.2 Rx Sensitivity Vs. Data rate



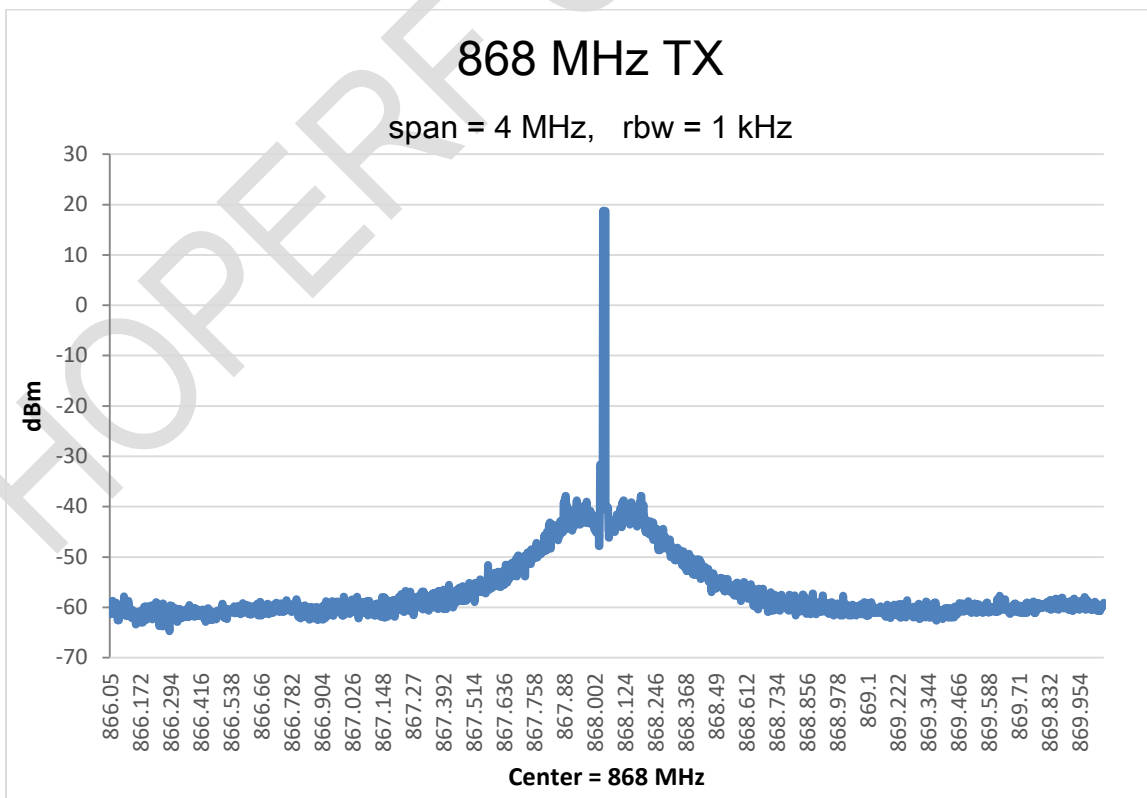
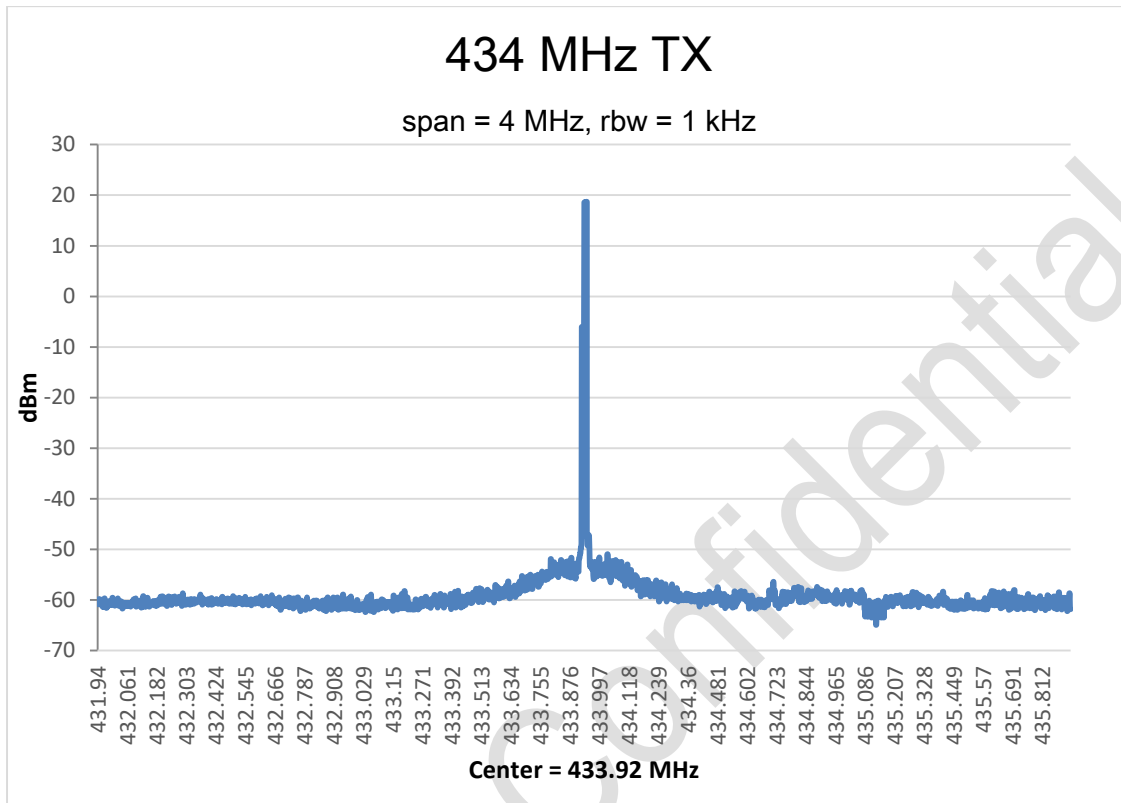
Test condition: Freq = 434 MHz, ppm = 10, BER <=0.1%

1.12.3 Tx Power Vs. Supply Power Voltage



Test condition: Freq = 434 MHz, 20 dBm matching network, Tx power with 3.3 V and 20 dBm

1.12.4 Tx Phase Noise



2 Pin Descriptions

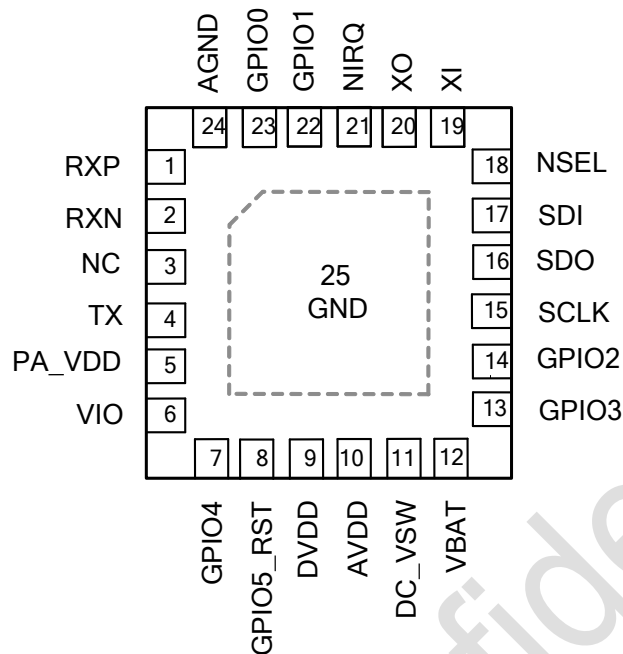


Figure 1. CMT2312A Pin Arrangement

Table 12. CMT2312A Pin Description

Pin #	Pin Name	I/O	Description
1	RXP	I	RF signal positive input
2	RXN	I	RF signal negative input
3	NC	O	Floating, not connected
4	TX	O	Tx output
5	PA_VDD	IO	PA VDD
6	VIO	IO	IO VDD
7	GPIO4	IO	Configurable, see <i>Table 17 CMT2312A GPIO</i> for more details.
8	GPIO5_RST	IO	Configurable, see <i>Table 17 CMT2312A GPIO</i> for more details.
9	DVDD	I	Digital VDD
10	AVDD	I	Analog VDD
11	DC_VSW	I	DCDC
12	VBAT	I	Analog VDD
13	GPIO3	IO	Configurable, see <i>Table 18 CMT2312A GPIO</i> for more details.
14	GPIO2	IO	Configurable, see <i>Table 18 CMT2312A GPIO</i> for more details.
15	SCLK	I	SPI clock.
16	SDO	O	SPI data output.
17	SDI	I	SPI data input.
18	CSB	I	Chip select bar
19	XI	I	32M Crystal circuit input.
20	XO	O	32M Crystal circuit output.
21	NIRQ	I	Configurable, see <i>Table 17 CMT2312A GPIO</i> for more details.
22	GPIO1	IO	Configurable, see <i>Table 17 CMT2312A GPIO</i> for more details.
23	GPIO0	IO	Configurable, see <i>Table 17 CMT2312A GPIO</i> for more details.

24	AGND	I	Analog GND.
25	GND	I	GND

3 Typical Application Schematic

3.1 Direct Tie Schematic Diagram

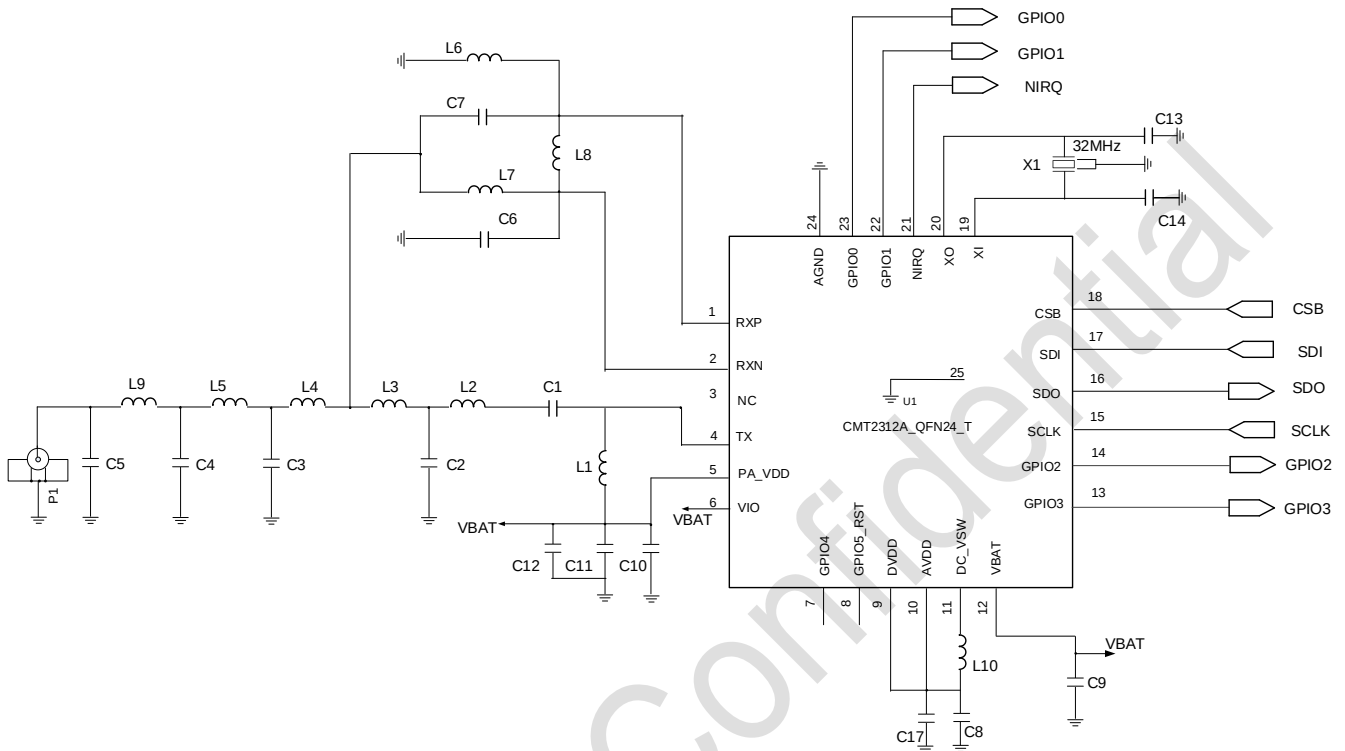


Figure 2. Direct Tie Application Schematic Diagram (DCDC enabled)

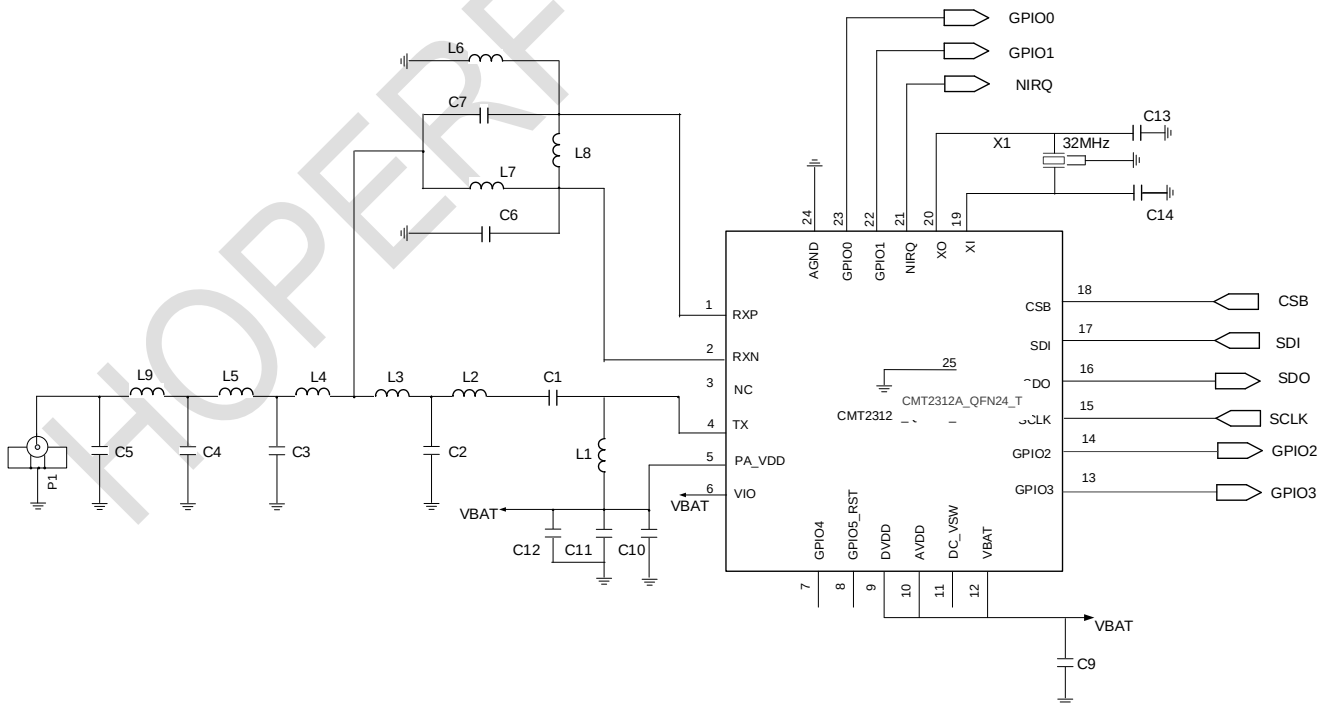


Figure 3. Direct Tie Application Schematic Diagram (DCDC disabled)

Table 3-1. 20 dBm Direct Tie Application BOM

No.	Descriptions	Component Values				Unit	Supplier
		315 MHz +20 dBm	433 MHz +20 dBm	868 MHz +20 dBm	915 MHz +20 dBm		
C1	±5%, 0402 NP0, 50 V	22	12	12	12	pF	
C2	±5%, 0402 NP0, 50 V	6.8	5.6	3.3	3.3	pF	
C3	±5%, 0402 NP0, 50 V	8.2	6.2	3.3	3.0	pF	
C4	±5%, 0402 NP0, 50 V	8.2	NC	NC	NC	pF	
C5	±5%, 0402 NP0, 50 V	NC	NC	NC	NC	pF	
C6	±5%, 0402 NP0, 50 V	5.6	3.9	1.8	1.8	pF	
C7	±5%, 0402 NP0, 50 V	5.6	3.9	1.8	1.8	pF	
C8	±5%, 0603 NP0, 50 V	2.2				uF	
C9	±5%, 0402 NP0, 50 V	1				uF	
C10	±5%, 0402 NP0, 50 V	220				pF	
C11	±5%, 0402 NP0, 50 V	100				nF	
C12	±5%, 0603 NP0, 50 V	4.7				uF	
C13	±5%, 0402 NP0, 50 V	NC				pF	
C14	±5%, 0402 NP0, 50 V	NC				pF	
C17	±5%, 0402 NP0, 50 V	100				nF	
L1	±5%, 0603 Chip Ceramic Inductor	220	180	100	100	nH	Sunlord
L2	±5%, 0603 Chip Ceramic Inductor	68	47	15	12	nH	Sunlord
L3	±5%, 0603 Chip Ceramic Inductor	56	39	15	12	nH	Sunlord
L4	±5%, 0603 Chip Ceramic Inductor	33	33	8.2	6.2	nH	Sunlord
L5	±5%, 0603 Chip Ceramic Inductor	47	33	8.2	6.2	nH	Sunlord
L6	±5%, 0603 Chip Ceramic Inductor	47	33	15	12	nH	Sunlord
L7	±5%, 0603 Chip Ceramic Inductor	47	33	15	12	nH	Sunlord
L8	±5%, 0603 Chip Ceramic Inductor	220	68	33	33	nH	Sunlord
L9	±5%, 0603 Chip Ceramic Inductor	33	NC	NC	NC	nH	Sunlord
L10	MPH252012C100MT, 10UH ±20%, Package 2520, DC Resistance 0.5 Ω, Saturation Current 0.5A	10				uH	Sunlord
X1	±10 ppm, SMD	32				MHz	EPSON
U1	CMT2312A RF receiver and transmitter					-	CMOSTEK

Table 3-2. 13 dBm Direct Tie Application BOM

No.	Descriptions	Component Values				Unit	Supplier
		315 MHz +13 dBm	433 MHz +13 dBm	868 MHz +13 dBm	915 MHz +13 dBm		
C1	±5%, 0402 NP0, 50 V	8.2	18	15	15	pF	
C2	±5%, 0402 NP0, 50 V	3.9	5.6	3.9	4.3	pF	
C3	±5%, 0402 NP0, 50 V	9.1	6.8	3.3	3.0	pF	
C4	±5%, 0402 NP0, 50 V	9.1	NC	NC	NC	pF	
C5	±5%, 0402 NP0, 50 V	NC	NC	NC	NC	pF	
C6	±5%, 0402 NP0, 50 V	5.6	3.9	1.8	1.8	pF	
C7	±5%, 0402 NP0, 50 V	5.6	3.9	1.8	1.8	pF	
C8	±5%, 0603 NP0, 50 V	2.2				uF	
C9	±5%, 0402 NP0, 50 V	1				uF	
C10	±5%, 0402 NP0, 50 V	220				pF	
C11	±5%, 0402 NP0, 50 V	100				nF	
C12	±5%, 0603 NP0, 50 V	4.7				uF	
C13	±5%, 0402 NP0, 50 V	NC				pF	
C14	±5%, 0402 NP0, 50 V	NC				pF	
C17	±5%, 0402 NP0, 50 V	100				nF	
L1	±5%, 0603Chip Ceramic Inductor	220	180	100	100	nH	Sunlord
L2	±5%, 0603Chip Ceramic Inductor	56	56	15	12	nH	Sunlord
L3	±5%, 0603Chip Ceramic Inductor	10	47	15	12	nH	Sunlord
L4	±5%, 0603Chip Ceramic Inductor	33	15	8.2	8.2	nH	Sunlord
L5	±5%, 0603Chip Ceramic Inductor	56	15	8.2	8.2	nH	Sunlord
L6	±5%, 0603Chip Ceramic Inductor	47	33	15	12	nH	Sunlord
L7	±5%, 0603Chip Ceramic Inductor	47	33	15	12	nH	Sunlord
L8	±5%, 0603Chip Ceramic Inductor	220	68	33	33	nH	Sunlord
L9	±5%, 0603Chip Ceramic Inductor	33	NC	NC	NC	nH	Sunlord
L10	MPH252012C100MT, 10UH ±20%, Package 2520, DC Resistance 0.5 Ω , Saturation Current 0.5A	10				uH	Sunlord
X1	±10 ppm, SMD	32				MHz	EPSON
U1	CMT2312A RF receiver and transmitter					-	CMOSTEK

4 Function Description

The CMT2312A is a high integrated mixed-signal transceiver, which employs a 32 MHz crystal to provide PLL reference frequency and digital clock with supporting of OOK, 2-(G)FSK, 4(G)FSK, G(MSK) modulation/demodulation as well as Direct and Packet Data processing modes.

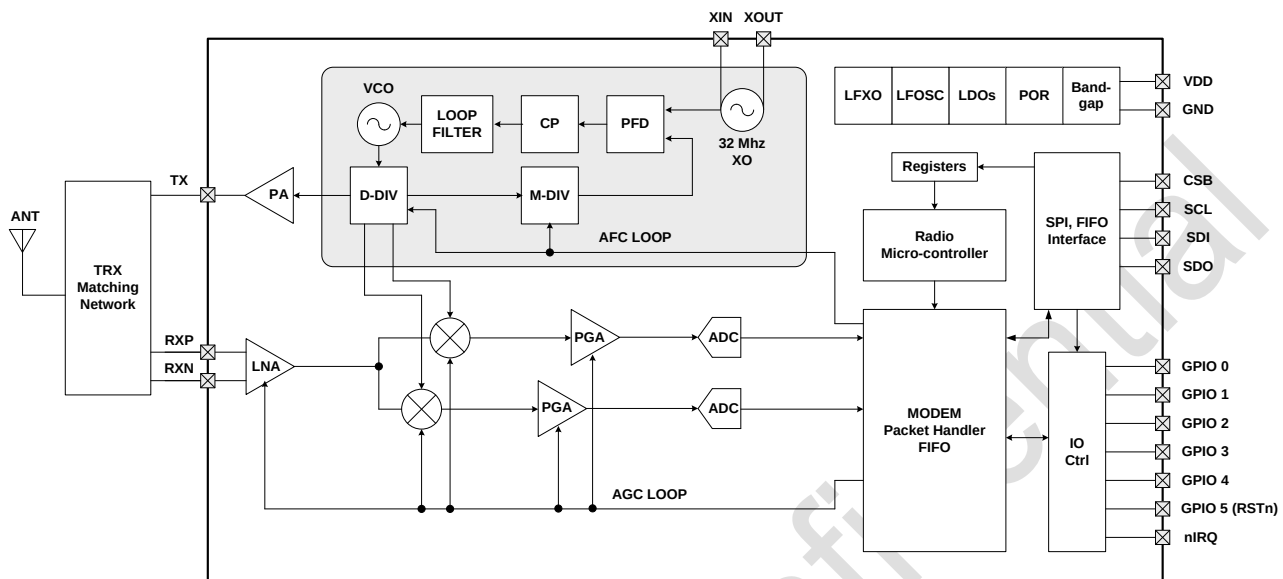


Figure 2. Functional Block Diagram

In receiver part, the chip applies LNA+MXR+PGA+ADC+PLL low-IF architecture to achieve the Sub-GHz wireless receiving function. In transmitter part, it applies PLL+PA architecture to achieve the Sub-1 GHz wireless transmitting function. The chip can achieve high-performance RF transmission and receiving with no need of extra RF switch components,

In the receiver system, the analog circuit mixes the RF signal to IF and converts the signal from analog to digital through ADC module, then outputs the two signal paths I/Q to the digital circuit for digital demodulation. The digital circuit is responsible for mixing the IF signal to zero-frequency (baseband) and performing a series of filtering and decision processing, meanwhile it applies AFC and AGC to control the analog circuit dynamically, and finally the original signal is demodulated. After demodulation, the signal is sent to the decoder for decoding and then filled into FIFO, or output to GPIO directly.

In the transmitter system, the digital circuit encodes and packets the data then sends it to the modulator (or send data directly to modulator without encoding). Then modulator will directly control the PLL and PA to have 2(G)FSK, 4(G)FSK or OOK, G(MSK) based modulation then transmit out the data.

The chip integrates a mini micro-controller (supporting supplier internal programming only), which controls and schedules various functions and operations like auto-ACK, auto-frequency-hopping, DUTY-CYCLE based low-power Tx and Rx, CSMA function, etc.

The chip provides SPI communication port. The external MCU can access registers to have various function configures, to control the chip or access FIFO.

4.1 Transmitter

The transmitter of CMT2312A is based on direct frequency synthesis technology. Its carrier is generated by a low-noise fractional-N frequency synthesizer. The modulated data is transmitted by an efficient single-ended power amplifier (PA). The output power can be read and written via the register, with configurable value ranging from -10 dBm to +20 dBm by a step of 1dB.

In OOK mode, when PA switches on and off quickly according to data transmitted, it will easily produce spectral spurs and glitches near the target carrier. Through the PA Ramping mechanism, it can help to minimize the spurs and glitches. In FSK mode, the transmitter supports to have Gaussian filtering on signals before transmission, namely GFSK, thus to make the transmitting spectrum more concentrated.

Users can design a PA matching network based on specific application requirements to optimize the transmission efficiency at the required output power. Typical application schematics and required BOMs are detailed in *Section 3 Typical Application Schematic*.

The transmitter can work in packet mode. In packet mode, data can be pre-loaded into the FIFO in STBY status and then transmitted along with other packet elements. In 4FSK mode, it only supports data transmitting from FIFO.

4.2 Receiver

An ultra-low-power, high-performance low-IF OOK/FSK receiver is built in the CMT2312A. Its processing steps are as follows: 1) the RF signal sensed by the antenna is amplified by the low-noise amplifier; 2) the signal is down-converted to the intermediate frequency by the quadrature mixer; 3) the signal is further amplified by the programmable amplifier; 4) The signal is sent to the digital domain through A/D convertor for digital demodulation processing. During power on reset (POR), each analog module is calibrated to the internal reference voltage. This allows the chip to keep its best performance at different temperatures and voltages. Baseband filtering and demodulation is accomplished by the digital demodulator. The AGC loop adjusts the system gain to optimize the performance parameters such as system linearity, selectivity and sensitivity.

Leveraging CMOSTEK's low-power design technology, the receiver in always-on mode consumes only a very low power. The periodic operating mode and wake-up function can further reduce the average power consumption of the system to satisfy applications with strict power consumption requirements.

Similar to the transmitter, the CMT2312A receiver can operate in direct mode and packet mode as well. In direct mode, the demodulator output data can be directly output through the DOUT pin of the chip. DOUT can be assigned to GPIO by configuration. In packet mode, the demodulator data output is sent to the data packet handler, get decoded then is filled into the FIFO, then MCU can read the FIFO through SPI interface.

4.3 Additional Functions

4.3.1 Power-On Reset (POR)

The power-on reset circuit detects the change of the VDD power supply and generate reset signal to reset the entire CMT2312A system. After the POR, the MCU must go through initialization process and re-configure the CMT2312A. There are two situations that will lead to the generation of POR.

The first case is when the supply voltage drops to 0.3V (+/-20%), the POR is reset to 0, and then the supply voltage is pulled up to 1.45V (+/-20%) at a speed greater than 1V/ms with POR changes from 0 to 1. As shown in the following figure:

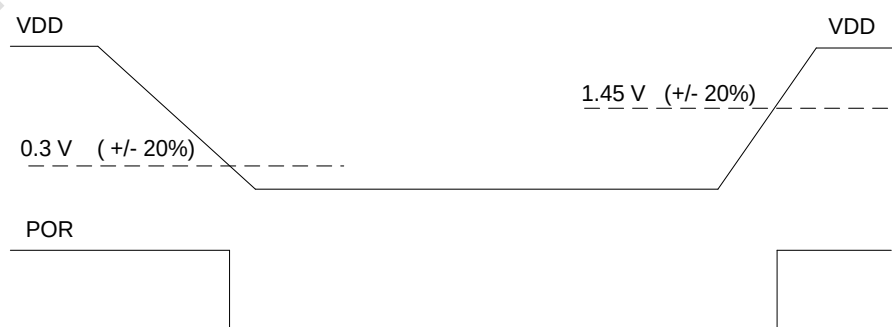


Figure 3. Sudden Decrease of VDD lead to POR Reset

The second scenario is a short-lived power supply mutation that causes the POR to reset. The triggering condition is that VDD plummets by 0.9V (+/-20%) in less than 2 us. Note that it monitors the decrease in VDD, not the absolute value of VDD. As shown in the figure below:

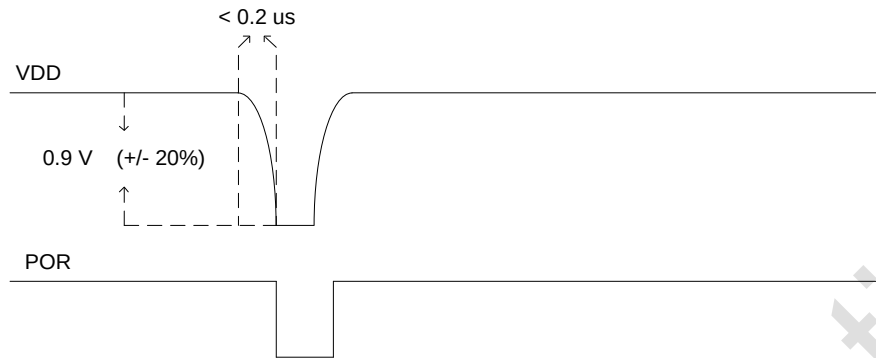


Figure 4. Rapid Decrease in Supply Voltage causes POR Reset

4.3.2 Crystal Oscillator

Crystal oscillators are used to provide a reference clock for the phase - locked loop and a system clock for digital modules. The differential load capacitance between XI and XO is equal to C_L . C_L should match the load capacitance value required by the crystal so that the crystal oscillates accurately at 32 MHz.

$$C_L = \frac{C_{onchip} + C_{off_chip} + C_{par}}{2}$$

C_{onchip} is the load capacitance connected from each end of the crystal provided internally for CMT2312A to the ground. The adjustable range can be configured through Xtal Cap Load on the RFPDK, which is 23 - 29pF with a step of approximately 190fF. $C_{offchip}$ is the load capacitance connected from the two ends of the externally connected crystal to the ground, and the customer can choose whether to add it. C_{par} is the parasitic capacitance from each end of the crystal to the ground, which is approximately 2 - 6pF. Here, it is recommended that users use a crystal oscillator with a 12pF load in combination with CMT2312A. In addition, the lower the ppm of the crystal, the better the performance of the receiver.

4.3.3 Temperature Compensated Crystal Oscillator (TCXO)

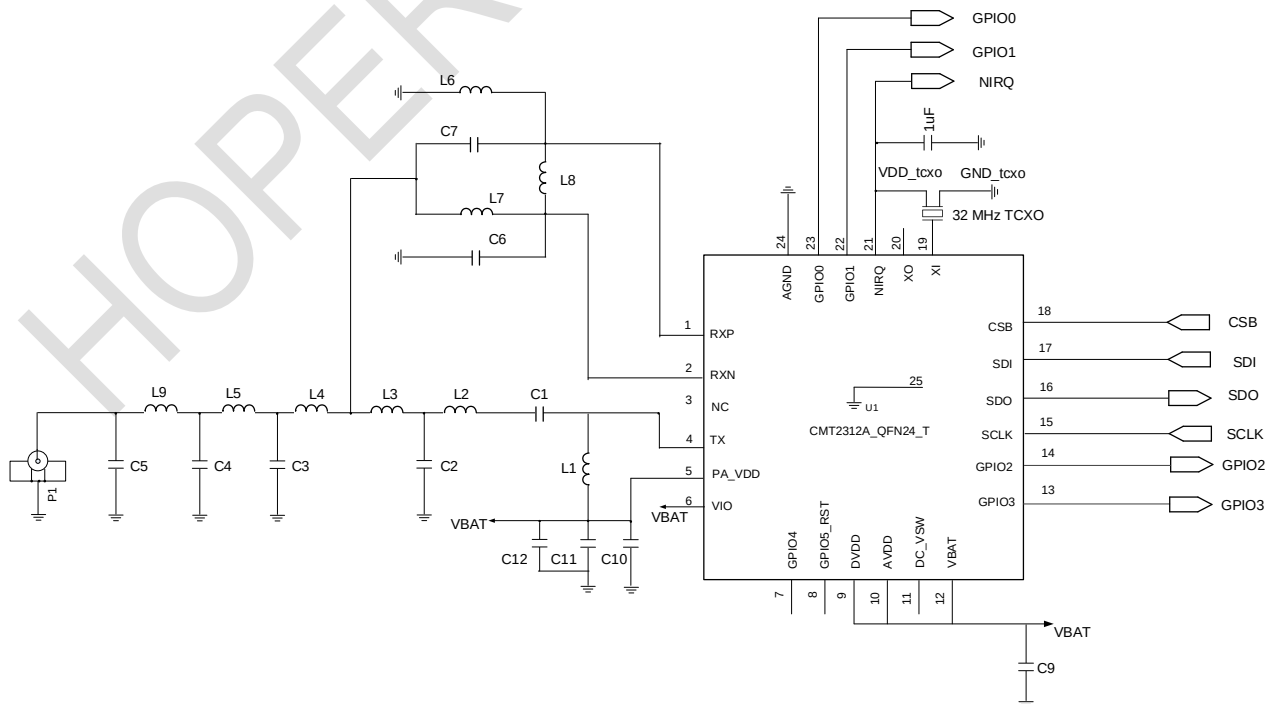


Figure 5. Application Schematic Diagram with TCXO

The CMT2312A allows users to use an external clock module, such as a temperature-compensated crystal oscillator (TCXO). The application schematic is shown in the figure above. The VDD pin of the TCXO is connected to the power supply by the NIRQ pin of the CMT2312A. Here, set NIRQ_SEL to 4, set the Output-Drive Capability on the RFPDK to 4mA, and add a 1uF filtering capacitor on the NIRQ PIN to achieve a good usage effect. The 32MHz clock of the TCXO is directly connected and output to the XI pin. In the IDLE and SLEEP states, the system will turn off the TCXO, and turn it on in other states.

At this time, the recommended peak-to-peak value of the clock signal is between 300mV and 700mV. Users can add a DC-blocking capacitor and a capacitor to ground at the XI-pin. By adjusting the ratio of these two capacitors, the effect of adjusting the input clock amplitude can be achieved. The use of TCXO will probably increase the overall current by 1.5 - 2.0mA. The precise clock (<2ppm) enables the CMT2312A to use a narrower receiving bandwidth, obtaining better sensitivity and the performance of anti-out-of-band interference.

4.3.4 Sleep Timer

The CMT2312A integrates a 32 kHz low-power oscillator (LPOSC). When this function is enabled, the timer periodically wakes the chip from sleep. When the chip operates in the periodic operation mode, the sleep time can be configured from 62.5 μ s to 8585740.288 s. Since the frequency of the low-power oscillator drifts with changes in temperature and voltage, it is automatically calibrated during the power-on phase and is periodically calibrated. These calibrations keep the frequency tolerance of the oscillator within $\pm 1\%$.

4.3.5 Low Battery Detection

The chip is equipped with a low - voltage detection function. This detection is carried out once every time the chip is tuned to a certain frequency. Frequency tuning happens when the chip jumps from the SLEEP/STBY state to the RFS/TFS/TX/RX state. The result can be read by the LBD_VALUE register.

4.3.6 Received Signal Strength Indicator (RSSI)

RSSI is used to evaluate the signal strength within the tuned channel, with a detection range from -127 dBm to 20 dBm. Users can select whether to output the RSSI value in real-time or latch the RSSI value at various stages of receiving data packets by configuring the RSSI - Detect Mode in RFPDK.

The CMT2312A allows users to set the detection threshold RSSI-CompareTH in the RFPDK. This value is compared with the detected RSSI value. If the detected RSSI value is greater than the threshold, 1 is output; otherwise, 0 is output. The comparison result can be output to the RSSI-VLD interrupt or used to assist the operation of the internal ultra-low power consumption (SLP) mode.

4.3.7 Phase Jump Detector (PJD)

PJD refers to Phase Jump Detector, that is, the phase jump detector. When the chip performs 2-FSK demodulation, it can be used to identify whether the received signal is noise or a useful signal by observing the jump characteristics of the received signal. OOK and 4-FSK demodulation do not support this function.

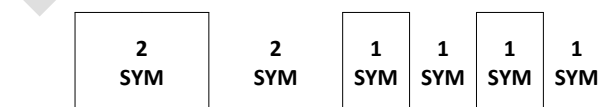


Figure 6. Received Signal Jump Diagram

The PJD mechanism regards input signal switching from 0 to 1 or from 1 to 0 as a phase jump. Users can configure PJD_WIN_SEL<1:0> to determine the number of detected jumps for the PJD to identify a wanted signal. As shown in the figure above, a total of 8 symbols are received, but the transition only occurs 6 times. Therefore, the number of transitions is not equal to the number of symbols. Only when receiving the preamble, the number of transitions is equal to the number of symbols. Generally speaking, the more times the PJD transitions occur, the more reliable the judgment result; the fewer times, the faster the completion. If the received time window is very small, then the number of detections needs to be reduced to meet the requirements of the window setting. Generally, when the number of transitions is 4 times, a relatively reliable detection effect can be achieved, that is, noise will not be misjudged as useful signals, and useful signals will not be undetected when they come.

Monitoring signal transitions essentially means monitoring whether the signal conforms to the expected data rate. While monitoring the transitions, PJD will also automatically internally monitor whether the Deviation of the signal conforms to the agreed value of a valid signal, and at the same time, it will judge whether the SNR exceeds 7dB. Then, by synthesizing the monitoring results of the data rate, Deviation, and SNR, if it is determined that the received signal is reliable, then 1 will be output; if it is noise or an interfering

signal, 0 will be output. This result can be configured to be output to the RSSIVLD interrupt or can be output to the implementation of the internal auxiliary ultra-low-power (SLP) receiving mode. In the direct-through mode, by setting the DOUT_MUTE register to 1, PJD can also be used to achieve FSK demodulation output muting.

The PJD technology is similar to the traditional Carrier Sensing (CS) technology, but it is more reliable than CS. When users combine RSSI monitoring with the PJD technology, they can accurately identify the idle state of the current channel.

4.3.8 Automatic Frequency Control (AFC)

The AFC function can help the receiver eliminate the frequency difference between TX and RX in a very short time when receiving signals, achieving higher sensitivity. The CMT2312A has top - notch AFC performance in the industry. Compared with other similar products, under the same bandwidth, the CMT2312A can identify a larger frequency difference and can eliminate the frequency deviation within a shorter time (8 - 10 symbols).

Since the frequency difference between TX and RX is usually caused by the frequency deviation of the crystals used by the two, the CMT2312A allows users to configure the tolerance value (unit: PPM) of the crystals used by TX and RX through the RFPDK, and configure the detection range of the AFC according to this tolerance value. At the same time, it ensures that the receiver uses the narrowest possible bandwidth. Due to the excellent performance of the AFC, the problem of the frequency deviation increasing over time due to crystal aging is better solved. The CMT2312A can support longer-term or more severe crystal aging, extending the product's service life.

4.3.9 Clock Data Recovery (CDR)

The basic task of a CDR system is to recover the clock signal that is synchronized with the symbol rate while receiving the data, not only for decoding inside the chip, but also for outputting the synchronized clock to GPIO for users to sample the data. So CDR's task is simple yet very important. If it has error between the recovered clock frequency and the actual symbol rate, it will cause data acquisition error, code error and decoding error during data receiving.

The CMT2312A provides three types of CDR systems as follows.

1. **COUNTING system** - The system is designed according to the case of relatively accurate symbol rate. If the symbol rate is 100% aligned, the unlimited length of 0 can be received continuously without error.
2. **TRACING system** - The system is designed according to the case of larger symbol rate error. It has tracking function, that is, it can automatically detect the Tx transmission symbol rate, and adjust the local symbol rate of RX quickly at the same time, so as to minimize the error between them. The system can withstand up to about 15.6% symbol rate error, an excellent level among similar products in the industry.
3. **MANCHESTER system** - This system evolves from the COUNTING system with the same basic features. The only difference is that the system is specially designed for Manchester code. It supports special processing to handle unexpected changes of TX symbol rate.

4.3.10 Fast Frequency Hopping

Manual frequency hopping means that, based on the basic frequency point configured using RFPDK, such as 433.92MHz, during the application process, the MCU can quickly switch to another frequency point by simply setting 1 to 2 registers. In multi-channel applications, it simplifies the operation of users constantly switching to new frequency points.

$$FREQ = \text{Base Freq} + 2.5 \text{ kHz} \times FH_OFFSET < 7:0 > \times FH_CHANNEL < 7:0 >$$

In general, users can configure FH_OFFSET<7:0> during the chip initialization process, and then in the application users can switch the channel by changing FH_CHANNEL<7:0>.

5 Chip Operation

5.1 SPI Interface

5.1.1 Read/Write Register

The chip communicates with the outside world through a 4 - wire SPI interface. The SPI is 4 - wire by default and can be configured to 3 - wire after power - on. The low - active CSB is the chip - select signal for accessing the register. SCLK is the serial port clock, and the maximum speed can reach 10MHz. Whether for the chip itself or the external MCU, data is sent on the falling edge of SCLK and collected on the rising edge. SDI is used for data input, and SDO is used for data output. In the 3 - wire mode, SDI is used for both data input and output, and SDO is idle. Both the address and data parts are transmitted starting from the MSB.

When accessing the register, the CSB should be pulled low. Then, an R/W bit is sent first, followed by a 7 - bit register address. After pulling the CSB low, the external MCU must wait for at least half a SCLK cycle before starting to send the R/W bit. After the falling edge of the last SCLK sent by the MCU, it must wait for at least half a SCLK cycle before pulling the CSB high.

It should be noted that for the 4 - wire write register operation in Figure 8, while writing data is input on SDI, the current value of the register (old-register read data) will be output on SDO. The MCU can decide whether to read this value according to its needs.

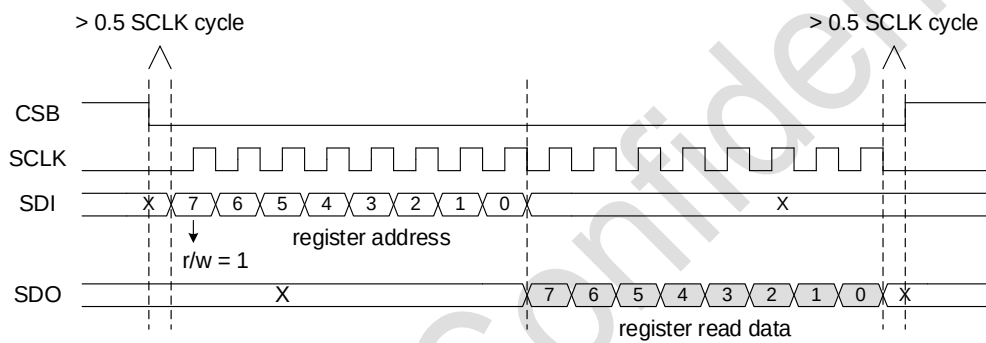


Figure 7. SPI (4-wire) Read Register Timing

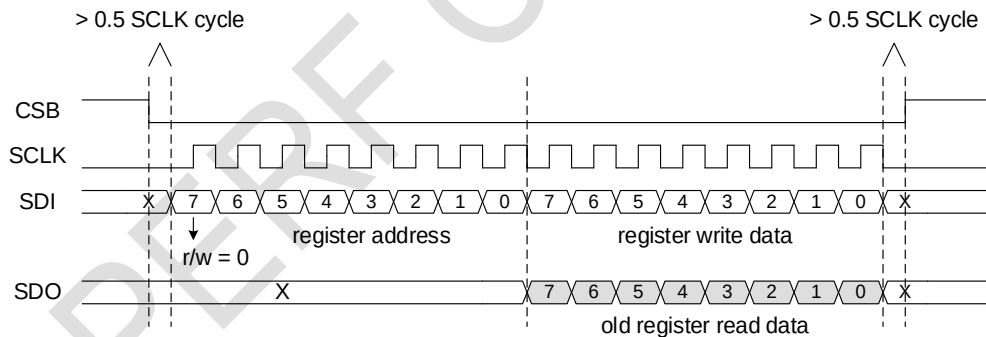


Figure 8. SPI (4-wire) Write Register Timing

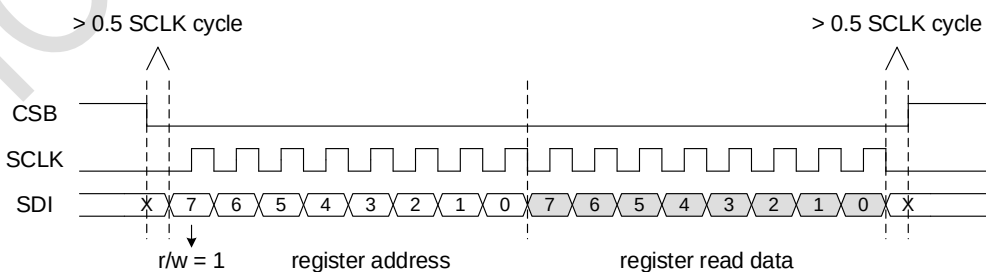


Figure 9. SPI (3-wire) Read Register Timing

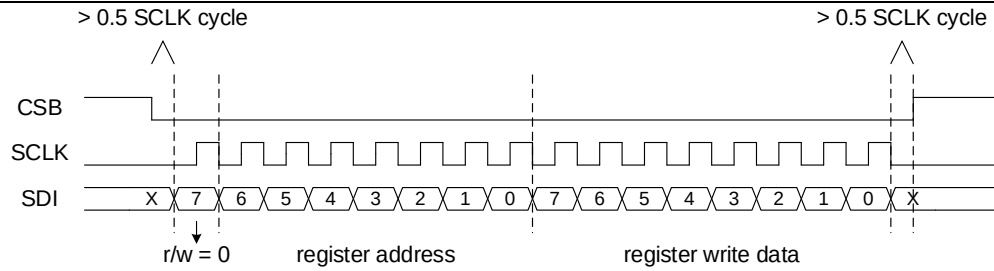


Figure 10. SPI (3-wire) Write Register Timing

For the read register operation of the 3 - wire interface, both the MCU and CMT2312A will perform the operation of switching the IO (SDIO) port between address 0 and data 7. At this time, the CMT2312A will switch the IO port from input to output, and the MCU will switch the IO port from output to input. Please note the position of the dotted line in the middle. At this time, it is highly recommended that the MCU switch the IO port to input before the falling edge of the SCLK is sent out; the CMT2312A will switch the IO to output only after seeing the falling edge. This avoids the situation where both set the SDIO to output simultaneously, resulting in an electrical conflict. For some MCUs, such a situation may cause them to reset or exhibit other abnormal behaviors.

5.1.2 Read/Write Register in Batch Mode (BURST)

In addition to the single-byte read and write register operations stated above, SPI can also support Burst read and write register operations. The BURST read and write operation is triggered by writing to the 0x7B address BRW PORT in Page 0. When the r/w bit is 0, a write register operation will be performed, and when it is 1, a read register operation will be carried out.

The BURST read/write can also be performed via 3-wire SPI. When 3-wire is used, outputting read data and inputting write data are both performed on the SDI pin. When using 4-wire, write data is input from SDI, and read data is output from SDO. The operation flow of the BURST read/ write is that: accessing the FIFO operation port at the address 0x7B first (the write/read bit inside determines whether it is a read or write operation), then followed by a stage of data reading/writing all along until users determine to complete the operation.

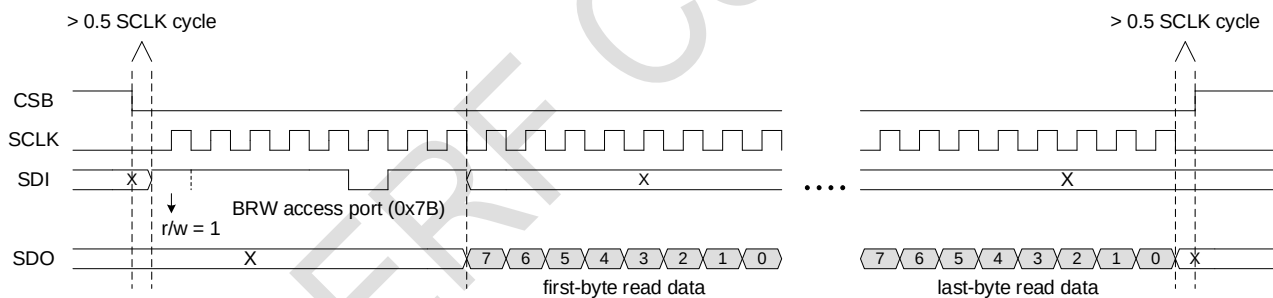


Figure 11. SPI (4-wire) BURST Read Register Timing

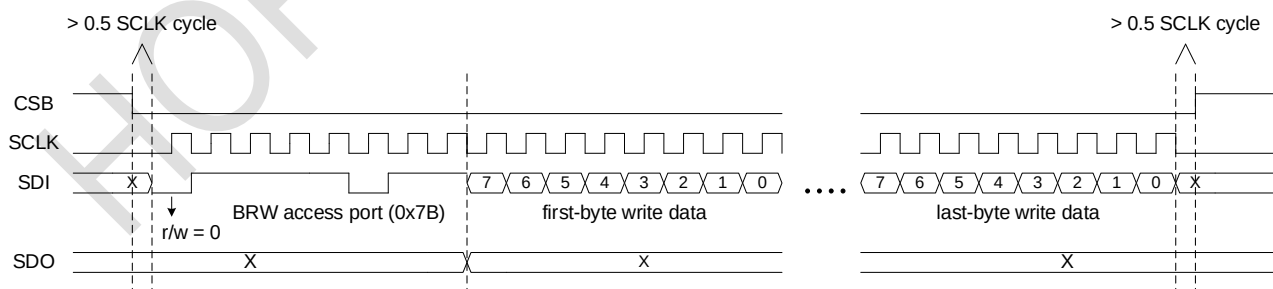


Figure 12. SPI (4-wire) BURST Write Register Timing

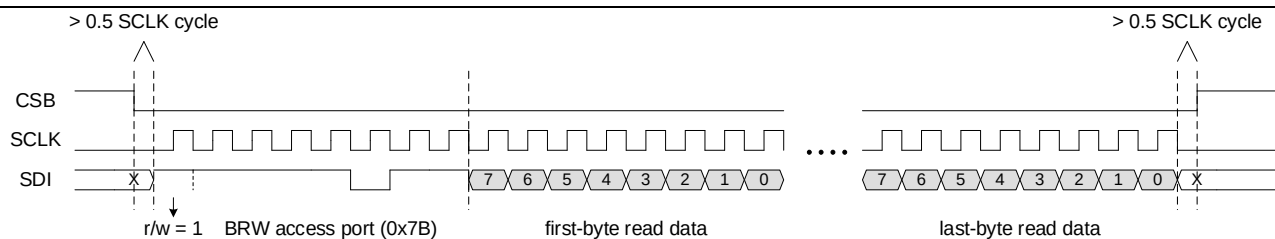


Figure 13. SPI (3-wire) BURST Read Register Timing

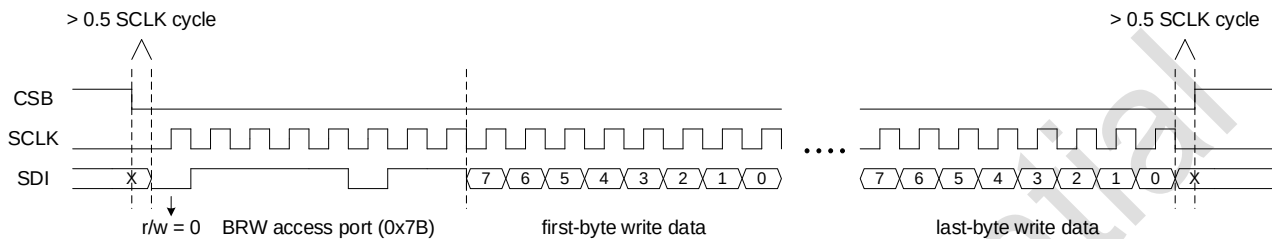


Figure 14. SPI (3-wire) BURST Write Register Timing

It should be noted that BURST read and write cannot cross Pages, that is, a single BURST can only be completed within one Page. The following lists the maximum address range for BURST read and write in each Page:

PAGE	Start Address	End Address	The number of addresses	Description
0	0x28	0x77	80	Configuration registers in Page0
1	0x00	0x6F	112	Configuration registers in Page1
2	0x00	0x3F	64	Frequency hopping table in Page2

Users can start from the 'starting address' each time within the address range of each Page and perform continuous BURST read and write operations on N addresses. The value of N should not cause the end address to exceed the address range. The common way of BURST read and write is to allow users to perform a quick one-time configuration before powering on. The content of the configuration register comes from the RFPDK tool, and the content of the frequency hopping channel table is designed by the users themselves.

5.2 FIFO

The CMT2312A provides two separated 128-byte FIFO by default. They are used for RX and TX respectively. RX FIFO is used to store received data in Rx mode while TX FIFO is used to store data to be transmitted in Tx mode. Users can also set FIFO_MARGE_EN to 1 to merge the two separated FIFO into one 256-byte FIFO, which is used both in TX and RX mode. By configuring the FIFO_RX_TX_SEL, it can indicate whether it is currently used as TX FIFO or RX FIFO. When the two FIFO are not merged, users can fill in the TX FIFO to prepare for the next transmission meanwhile the RX FIFO is being filled in, thus to save system operating time.

The CMT2312A provides two independent 128-byte FIFOs by default, which are used for RX and TX respectively and are independent of each other. The RX-FIFO is used to store received data in the RX mode, and the TX FIFO is used to store data to be transmitted in the TX mode. Users can also set FIFO_MARGE_EN to 1, so that the two FIFOs are combined into a 256-byte FIFO, which can be used in both TX and RX. The FIFO_RX_TX_SEL is configured to indicate whether it is used for TX or RX. If the merger is not used, when the 128-byte RX-FIFO is filled, the user can fill the 128-byte TX FIFO for the next transmission at the same time to save system operation time.

The FIFO can be accessed through the SPI interface. Users can clear the FIFO by setting the TX_FIFO_CLR/RX_FIFO_CLR bits. Additionally, users can set FIFO_RESTORE to re - transmit the previously filled data without the need to refill the data.

Users can configure PD_FIFO to control whether the FIFO saves its content in the SLEEP state. PD_FIFO= 0 means that the FIFO can save its content in the SLEEP state, but it will consume a leakage current of about 200 nA.

5.2.1 FIFO Read Operation

When the MCU needs to access the FIFO, it is necessary to configure some registers first to set the read/write mode of the FIFO and other working modes. The following figure shows the read/write timing diagram after the mode is determined. The operation of the FIFO is triggered by writing to the address 0x7A of Page0. When the r/w bit is 0, a write FIFO operation will be performed, and when it is 1, a read FIFO operation will be performed.

The read and write operations of FIFO can also be performed using a 3 - wire SPI. When using 3 - wire, both the output of read data and the input of write data are carried out on the SDI pin. When using 4 - wire, write data is input from SDI, and read data is output from SDO. The operation process of FIFO is to first access the FIFO operation port at address 0x7A. The read - write bit contained in it determines whether the subsequent operation is a write data or a read data operation. After that, it is the data stage of continuous reading or writing, and the user decides when to complete the operation.

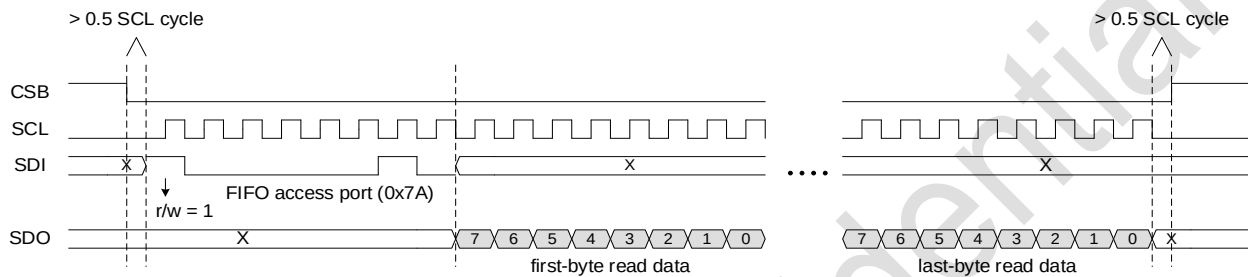


Figure 15. SPI (4-wire) Read FIFO Timing

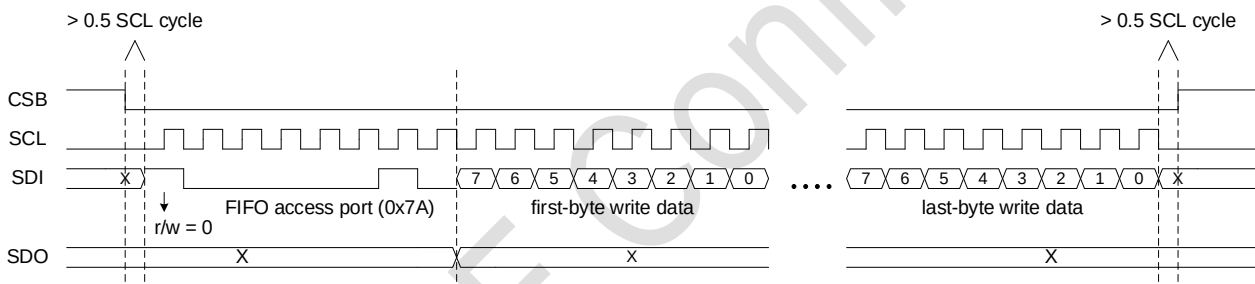


Figure 16. SPI (4-wire) Write FIFO Timing

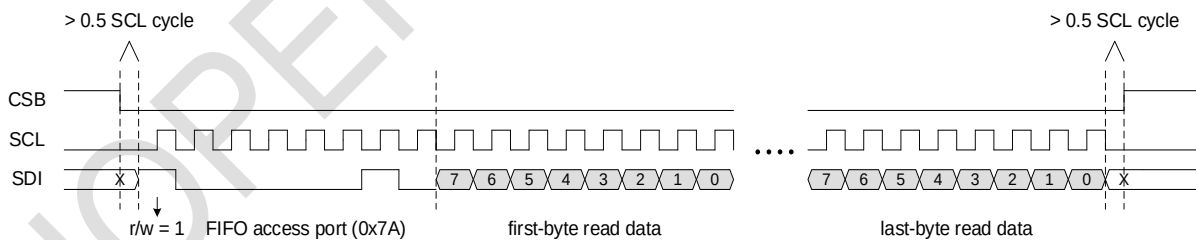


Figure 17. SPI (3-wire) Read FIFO Timing

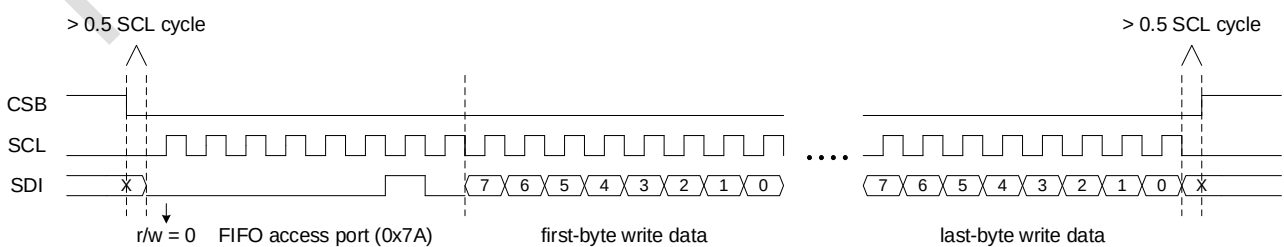


Figure 18. SPI (3-wire) Write FIFO Timing

5.2.2 FIFO Associated Interrupt

The CMT2312A provides a rich set of FIFO - related interrupt sources as an auxiliary means for the efficient operation of the chip. The FIFO interrupt timing related to Rx and Tx is shown in the following figure.

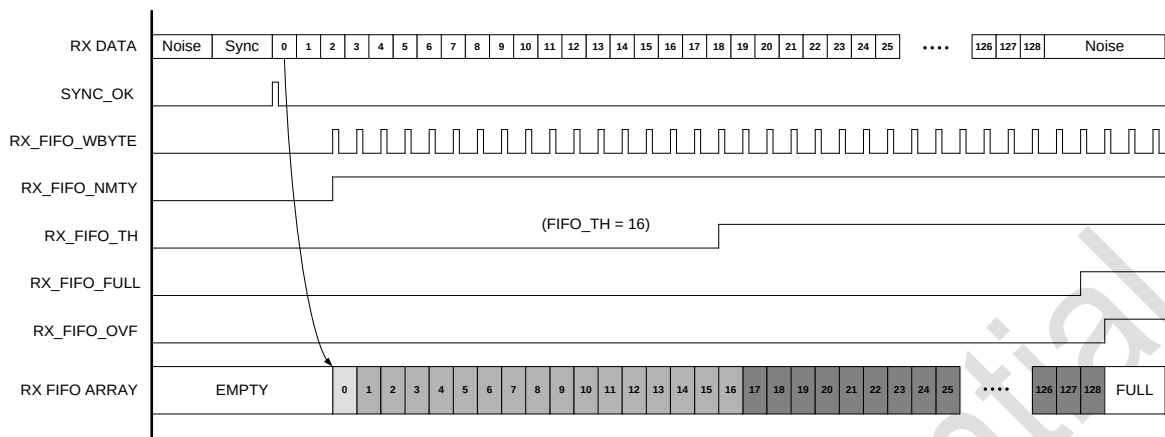


Figure 19. CMT2312A RX FIFO Interrupt Timing

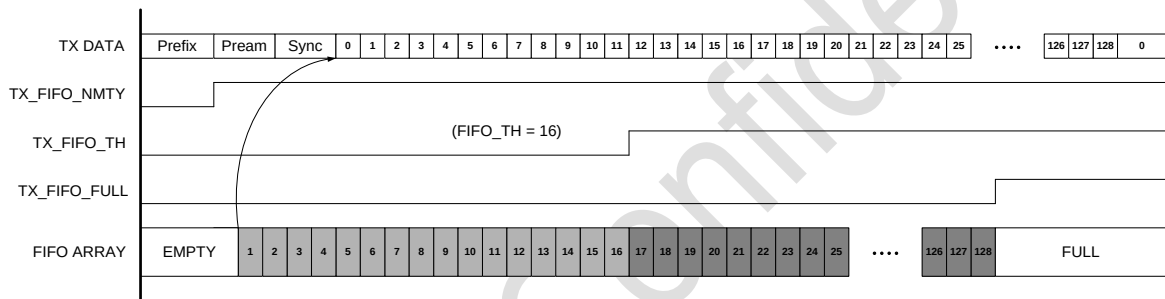


Figure 20. CMT2312A TX FIFO Interrupt Timing

5.3 Operation State, Timing and Power Consumption

5.3.1 Startup Timing

After VDD is turned on for the chip, the POR will be released after waiting for about 1 ms. However, the chip will remain in the IDLE state and not perform any actions. After the user sends the power_up command, the chip will start the power - on process and perform calibration for each module. After the chip completes the calibration, it will stay in the SLEEP state. At any time, as long as a reset is carried out (including POR, hard reset, and soft reset), the chip will return to the IDLE state and wait for the user to send the power_up command again.

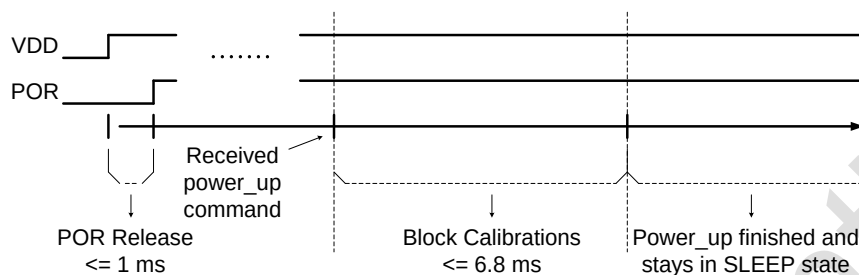


Figure 21. Power-on Timing

5.3.2 Operation State

The CMT2312A supports 7 operation states: IDLE, SLEEP, STBY, RFS, RX, TFS and TX, as shown in the figure below.

Table 13. Status and the Corresponding Enable Module of CMT2312A

State	Binary code	Switch ommand	Active Modules	Optional Moduels
IDLE	0x00	soft_rst	SPI, POR	None
SLEEP	0x81	go_sleep	SPI, POR	LFOSC, FIFO, Sleep Timer
READY	0x82	go_ready	SPI, POR, XTAL, FIFO	None
RFS	0x84	go_rfs	SPI, POR, XTAL, PLL, FIFO	None
TFS	0x88	go_tfs	SPI, POR, XTAL, PLL, FIFO	None
RX	0x90	go_rx	SPI, POR, XTAL, PLL, LNA+MIXER+ADC, FIFO	RX Timer
TX	0xA0	go_tx	SPI, POR, XTAL, PLL, PA, FIFO	None

The time for switching between various states is listed in the table below.

Table 14. State Switching Time

Source State	Destination State					
	SLEEP	READY	RFS	RX	TFS	TX
SLEEP		660 us	770 us	820 us	770 us	820 us
READY	Immediately		110 us	160 us	110 us	160 us
RFS	Immediately	Immediately		20 us	Cannot switch	Cannot switch
RX	Immediately	Immediately	Immediately		Cannot switch	160 us
TFS	Immediately	Immediately	Cannot switch	Cannot switch		20 us
TX	Immediately	Immediately	Cannot switch	160 us	Immediately	

Notes:

In Direct mode, if the chip is transmitting and a switching command is received, it will immediately exit the TX state.

In Packet mode, if the chip is transmitting, it must complete the current transmission before it can exit the TX state.

from SLEEP to TFS, the time for crystal startup and stabilization needs to be added. Switching to TFS from other states is completed immediately.

■ RX State

In RX, all modules related to the receiver will be turned on. It only takes 20us to switch from RFS to RX. When switching from STBY to RX, an additional 160us of PLL calibration and stabilization time is required. When switching from SLEEP to RX, the crystal startup and standard stabilization time need to be added. In TX, the go_switch command can be sent to quickly switch to RX. Whether the frequency points set for TX and RX are the same or not, it is necessary to wait for 160us of PLL recalibration and stabilization time before the switch can be successful.

■ TX State

In TX, all modules related to the transmitter will be turned on. It only takes 20us to switch from TFS to TXX. When switching from STBY to TX, an additional 160us of PLL calibration and stabilization time is required. When switching from SLEEP to TX, the time for the crystal to start up and stabilize needs to be added. In RX, the go_switch command can be sent to quickly switch to TX. Whether the frequency points set for RX and TX are the same or not, it is necessary to wait for 160us of PLL recalculation and stabilization time to complete the switch successfully.

5.4 GPIO and Interrupt

The CMT2312A has 7 GPIOs (GPIO0 - GPIO5 and NIRQ), and each GPIO can be configured as different input or output. The CMT2312A has 3 interrupt ports (INT1, INT2, INT3), which can be configured to different GPIO outputs.

Table 15. CMT2312A GPIO

Pin #	Name	I/O	Function
23	GPIO0	IO	Can be configured as: DOUT, INT1, INT2, INT3, DCLK, TRX_SWT.
22	GPIO1	IO	Can be configured as: DCLK, INT1, INT2, DOUT, TRX_SWT.
14	GPIO2	IO	Can be configured as: INT1, INT2, INT3, DCLK, DOUT, ANTD1.
13	GPIO3	IO	Can be configured as: INT1, INT2, DCLK, DOUT, ANTD2.
7	GPIO4	IO	Can be configured as: DOUT, INT1, INT2, DCLK, CLK0, LFCLK0.
8	GPIO5	IO	Can be configured as: RSTn, INT1, INT2, DOUT, DCLK.
21	NIRQ	IO	Can be configured as: INT1, INT2, DCLK, DOUT, TCXO.

The interrupt mapping is listed in the table below. As INT1 and INT2 have the same mapping, it takes INT 1 as an example in the table below.

Table 16. CMT2312A Interrupt Mapping

Name	INT1_SEL	Interrupt Descriptions	Clearing Method
INT_MIX	000000	The combination is interrupted. If any of the following interrupts is valid, INT_MIX will be valid.	Auto/By MCU
ANT_LOCK	000001	The antenna completes the locking interrupt after the antenna diversity function is running.	By MCU
RSSI_PJD_VALID	000010	A valid interruption of the combination of RSSI and/or PJD.	Auto
PREAM_PASS	000011	Interrupt indicating successful receipt of Preamble.	By MCU
SYNC_PASS	000100	Interrupt indicating successful receipt of Sync Word.	By MCU
ADDR_PASS	000101	Interrupt indicating successful receipt of Addr.	By MCU
CRC_PASS	000110	Interrupt indicating successful receipt and CRC check being passed.	By MCU
PKT_OK	000111	Interrupt indicating receipt of an entire and correct packet.	By MCU
PKT_DONE	001000	It indicates that the current data packet has been completely received. There are the following four situations: 1. The entire data packet is received completely and the data packet is correct. 2. Manchester decoding error, and the decoding circuit restarts automatically. 3. NODE-ID reception error, and the decoding circuit restarts automatically. 4. Signal conflict is detected. The decoding circuit does not restart automatically and waits for the MCU to handle it.	By MCU
SLEEP_TMO	001001	Interrupt indicating SLEEP timer timeout.	By MCU
RX_TMO	001010	Interrupt indicating RX timer timeout.	By MCU
RX_FIFO_NMTY	001011	Interrupt indicating RX FIFO being empty.	Auto
RX_FIFO_TH	001100	Interrupt indicating the unread content of RX FIFO exceeding FIFO TH.	Auto
RX_FIFO_FULL	001101	Interrupt indicating RX FIFO being full	Auto
RX_FIFO_WBYTE	001110	Interrupt generated every time a BYTE is written into RX FIFO, i.e., it is a pulse.	Auto

Name	INT1_SEL	Interrupt Descriptions	Clearing Method
RX_FIFO_OVF	001111	Interrupt indicating RX FIFO being overflow	Auto
TX_DONE	010000	Interrupt indicating TX completion.	By MCU
TX_FIFO_NMTY	010001	Interrupt indicating TX FIFO not being empty.	Auto
TX_FIFO_TH	010010	Interrupt indicating the unread content of TX FIFO exceeding FIFO TH.	Auto
TX_FIFO_FULL	010011	Interrupt indicating TX FIFO being full.	Auto
STATE_IS_READY	010100	Interrupt indicating that the current state is READY.	Auto
STATE_IS_FS	010101	Interrupt indicating that the current state is RFS or TFS.	Auto
STATE_IS_RX	010110	Interrupt indicating that the current state is RX.	Auto
STATE_IS_TX	010111	Interrupt indicating that the current state is TX.	Auto
LBD_STATUS	011000	Interrupt indicating that low voltage detection being active (VDD is lower than the set TH).	By MCU
API_CMD_FAILED	011001	Interrupt indicating API command execution error.	By MCU
API_DONE	011010	Interrupt indicating API command completion.	By MCU
TX_DC_DONE	011011	Interrupt for Duty Cycle TX mode complete	By MCU
ACK_RECV_FAILED	011100	Interrupt indicating ACK receiving failure.	By MCU
TX_RESEND_DONE	011111	Interrupt for repeated TX complete	By MCU
NACK_RECV	011110	Interrupt indicating receipt of NACK.	By MCU
SEQ_MATCH	011111	Interrupt indicating successful serial number matching.	By MCU
CSMA_DONE	100000	Interrupt for CSMA complete	By MCU
CCA_STATUS	100001	Signal channel sensing interrupt.	By MCU

The default interrupt is active at 1, but by setting the bit of the INT_POLAR register to 1, all interrupts become active at 0. Below, taking INT1 as an example, there are control and selection diagrams for two interrupt sources with different properties. For control and mapping, INT1 and INT2 are the same and can be mapped to any GPIO. The source of INT3 is only INT_MIX and can only be mapped to GPIO0 and GPIO2. When using it, users can choose to use INT_MIX to map all interrupt sources to the interrupt port and identify which interrupt is active by querying the interrupt flag; or they can directly map a certain interrupt source to the interrupt port.

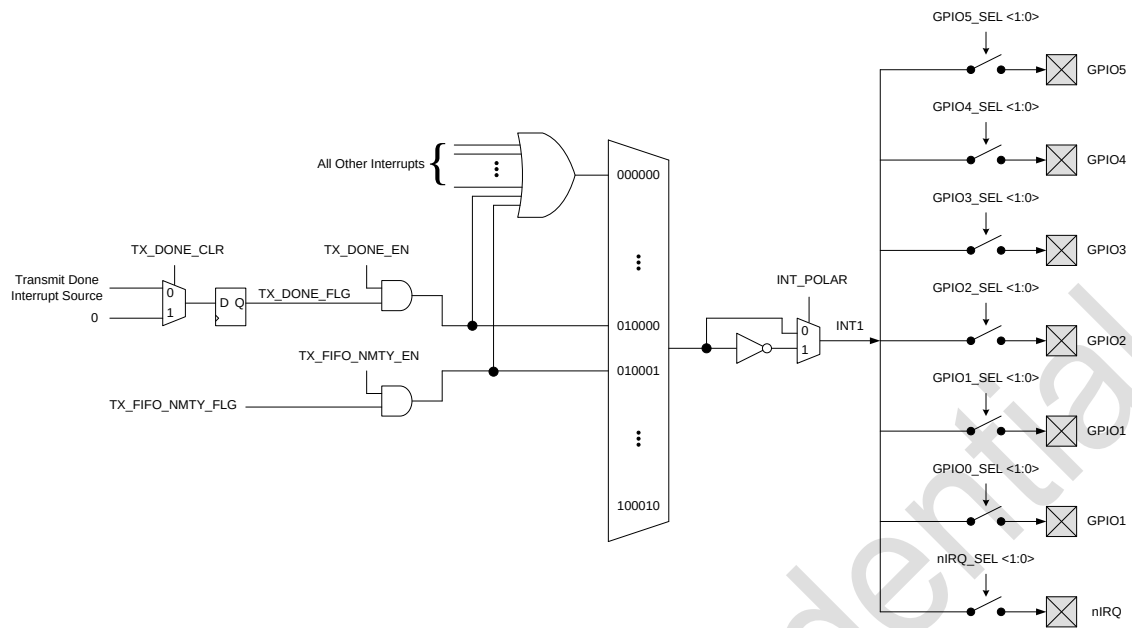


Figure 23. CMT2312A INT1 Interrupt Mapping Diagram

6 Data Packet and Packet Handler

Data Mode refers to the mode through which the external MCU inputs transmitted data or acquires received data. The CMT2312A supports two modes: direct-through receive mode and packet mode. The differences are as follows...

- Direct Rx Mode - In RX mode, only preamble and sync detection are supported, and the FIFO does not work.
- Packet Mode – it supports all packet formats and FIFO works.

6.1 Direct Rx Mode

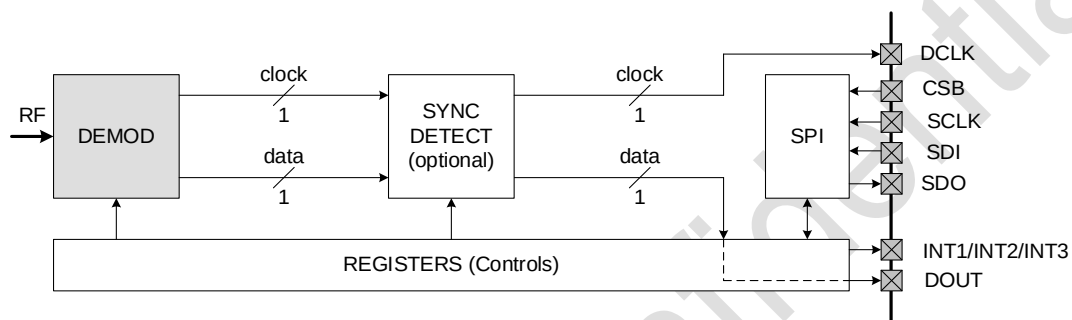


Figure 24. Direct Mode Data Path

Rx processing

In direct mode, data is sent directly from the output of the demodulator to the external MCU via DOUT. DOUT can be set to GPIO1 - GPIO5 and NIRQ. For the MCU, the typical RX working sequence in the direct pass - through mode is as follows:

- 1 Configure which GPIO to use.
- 2 Configure DATA_MODE = 0.
- 3 Send the go_rx command.
- 4 Capture the data from DOUT continuously.
- 5 Send the go_sleep/go_stby/go_rfs command to complete receiving and save power at the same time.

6.2 Packet Mode

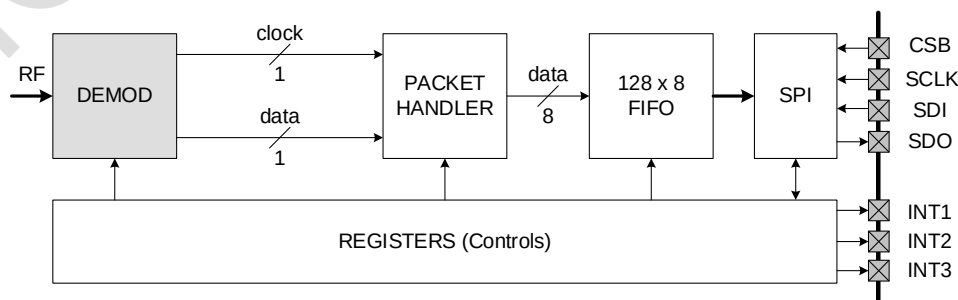


Figure 25. Packet Mode Data Path

The CMT2312A adopts unified TX and RX configuration, supports typical and more flexible packet formats, and can be divided into three types: variable packets (Length is before the Node ID), variable packets (Length is after the Node ID), and fixed packets. Every part of the data packet supports highly flexible configuration.

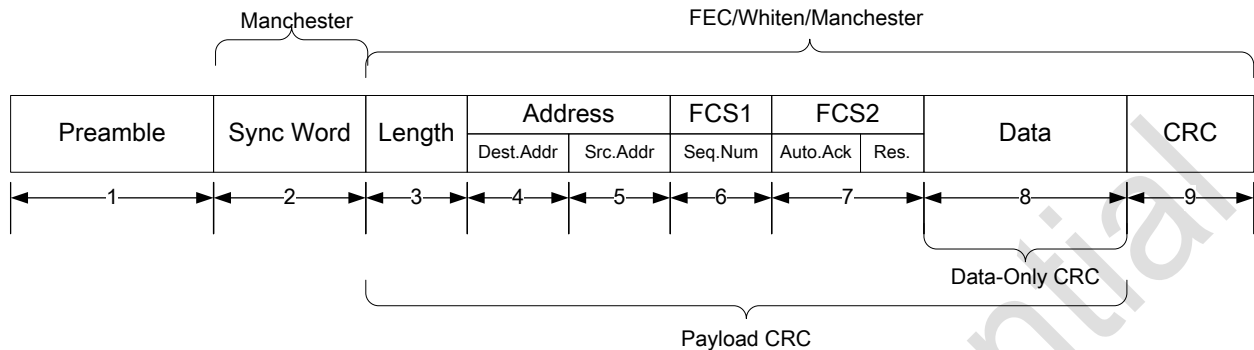


Figure 26. CMT2312A Packet Format

6.2.1 Rx processing

In the packet mode, the data output from the demodulator is first transferred to the packet processor for decoding, and then filled into the FIFO. The packet processor provides various decoding engines and options for judging the data validity, which can reduce the resources of the user's MCU. For the MCU, the typical Rx working sequence in the packet mode is as follows.

1. Configure which GPIO to use
2. Select the interrupts to be output.
3. Send go_rx command.
4. Read RX FIFO according to related interrupt status.
5. Send go_sleep/go_stby/go_rfs command to complete receiving and save power at the same time.
6. Clear interrupts.

6.2.2 Tx processing

In packet mode, the MCU must pre-load the FIFO with data in READY and TFS states, and then send the go_tx command to send the data out. The typical Tx packet mode control sequence for the MCU is shown as below:

1. Configure which GPIO to use
2. Send go_ready and fill data into FIFO.
3. Select the interrupts to be output.
4. Send go_tx command.
5. Write data into the TX-FIFO in the corresponding interrupt state.
6. After sending N data packets, it will automatically exit to the selectable state without the need to send commands.

The FIFO, packet processing mechanism, and related interrupt resources of CMT2312A are very rich, enabling seamless communication with the vast majority of similar products. For specific details, please refer to the configuration interface of RFPDK.

7 Low Power Operating

7.1 Duty Cycle Operating Mode

The CMT2312A configures relevant registers to enable the Tx and Rx of the chip to operate in the Duty - Cycle operation mode, thereby saving the chip's power consumption.

The Duty - Cycle mode of RX can be divided into the following five modes.

1. Fully manual mode.
2. Automatically wake-up from sleep and switch to manual control.
3. Automatically wake up from SLEEP, automatically enter RX, and manually exit RX.
4. Automatically wake up from SLEEP, manually enter RX, and automatically exit RX.
5. Fully automatic control.

The Duty-Cycle of TX can be divided into the following three modes.

1. Manually enter TX and automatically exit TX.
2. Automatically wake up from SLEEP, manually enter TX, and automatically exit TX.
3. Fully automatic transmission.

7.2 Super-low Power (SLP) Rx Mode

The CMT2312A provides a series of options that can help users achieve ultra-low power (SLP - Super Low Power) reception under different application requirements. These options will only take effect when RX_TIMER_EN is set to 1, that is, when the RX timer is valid. The core of SLP reception is how to minimize the RX time when there is no signal and appropriately extend the RX time for reception when there is a signal, ultimately achieving the effect of minimizing power consumption and stable reception.

Traditional short - range wireless transceiver systems generally implement low - power transceiving with the following basic scheme. The CMT2312A is also compatible with this scheme and, on this basis, has expanded 13 more power - saving schemes. First, let's introduce the most basic scheme at the following section.

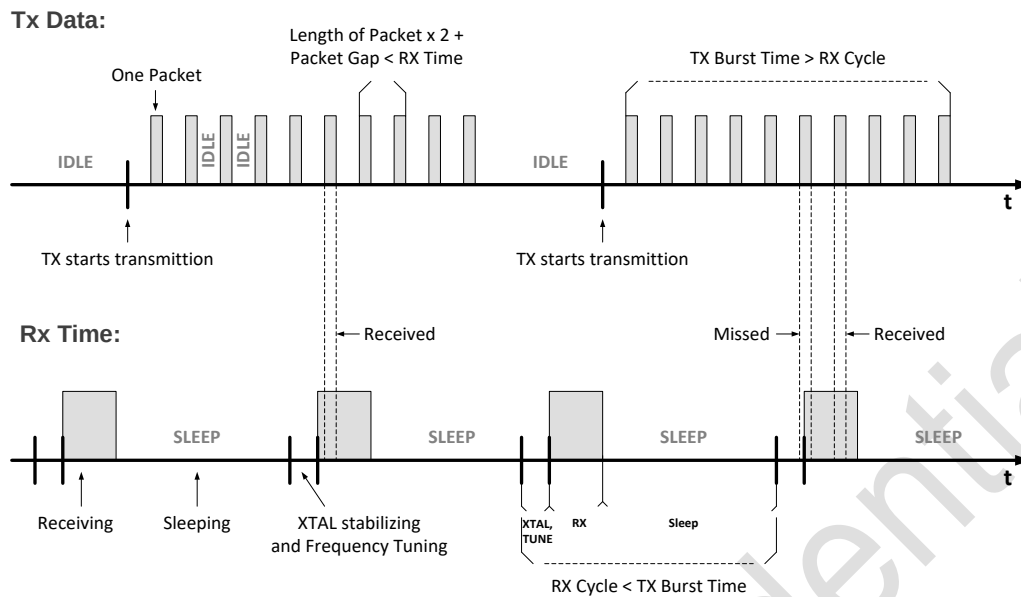


Figure 27. Basic Low-power Transceiver Scheme

The traditional low-power transceiver solution and the 13 extended low-power solutions based on it are listed in the table below.

Table 17. Low-power Rx Mode

No.	Rx Extended Methods	Rx Extended Condition
0	If the configuration is 0, no extension will be done, and it will leave RX when the T1 timing ends.	None
1	Once the detection conditions are met in T1, leave T1 and transfer control to the MCU.	RSSI_VLD is valid.
2		PREAM_OK is valid.
3		RSSI_VLD and PREAM_OK are valid at the same time.
4	As long as a valid RSSI is detected in T1, exit T1 and remain in RX state until the RSSI no longer meets the requirements, at which point exit RX.	RSSI_VLD is valid.
5	Once the detection conditions are met in T1, switch to T2. After the timing of T2 ends, exit RX.	RSSI_VLD is valid
6		PREAM_OK is valid
7		RSSI_VLD and PREAM_OK are valid at the same time.
8		Anyone of PREAM_OK or SYNC_OK is valid.
9		Anyone of PREAM_OK or NODE_OK is valid.
10		Anyone of PREAM_OK or SYNC_OK or NODE_OK is valid.
11	Once the detection condition is met in T1, switch to T2. Once SYNC is detected in T2, exit T2 and transfer control to the MCU. Otherwise, exit RX after the T2 timer ends.	RSSI_VLD is valid.
12		PREAM_OK is valid.
13		RSSI_VLD and PREAM_OK are valid at the same time.

Notes:

The signal source of RSSI_VLD can be selected as the comparison result of RSSI, or the monitoring result of PJD, or both are valid at the same time.

The T1 and T2 mentioned in the above table refer to the available registers or the RXT1 and RXT2 time windows set by RFPDK. The source of RSSIVLD can be the comparison result of RSSI or the result of phase detection (PJD).

7.3 Rx Automatic Frequency Hopping (RX AUTO HOP)

When the chip enters a channel for receiving, if no successful events are detected in the corresponding mode, the chip will return to READY or RFS state according to `FREQ_SW_STATE`, then automatically configure the channel value in the next byte in the hopping table then re-enter Rx state for receiving. If the automatic switched state is READY, the PLL frequency will be re-calibrated before re-entering RX with corresponding calibration time consumed; if the automatic switched state is RFS, the PLL frequency will not be re-calibrated, thus the switching time is shorter. If the interval between any two frequency points in the frequency hopping table exceeds 2MHz, it is recommended to configure `FREQ_SW_STATE` to the READY state, for it is necessary to recalibrate the PLL in this case.

If successful events are detected, the chip will auto-hop and wait for the further processing from the external MCU or switch to the pre-configured state. If successful events are detected when the entire table hopping is finished, the chip will hop from the start of the table if `RX_HOP_PERSIST` is configured as 1, or else return to the state specified in `RX_EXIT_STATE` if `RX_HOP_PERSIST` is configured as 0 to finish the Rx-auto-hopping operation. The destination frequency point calculation is as follows.

$$\text{FREQ} = 1\text{K} \times \text{FREQ_SPACE}<7:0> \times \text{FREQ_CHANL}<7:0>$$

In the calculation above, `FREQ_SPACE<7:0>` is an independent register, `FREQ_CHANL<7:0>` is configured through 64 registers with each one stores a single channel value.

The register `REQ_CHANL_LV` of `RX_Auto_Hop` mode lets users know which channel currently is used. Moreover, the register `FREQ_DONE_TIMES` is to let users read the number of currently finished hopping.

The `RX_Auto_Hop` mode needs to use `RX T1 TIMER` and `RX T2 TIMER`, therefore `TIMER_RX_EN` must be set to 1. `RX T1 TIMER` is configured through `TIMER_M_RX_T1` and `TIMER_R_RX_T1` and the actual timer time is calculated according to the timer formula. `RX T2 TIMER` is configured through `TIMER_M_RX_T2` and `TIMER_R_RX_T2` and the actual timer time is calculated according to the formula.

`CCA_INT_SEL` is used to determine whether `RSSI_PJD_VLD` is judged through `RSSI` or `PJD`; `RX_EXIT_STATE` is used to determine which state to be returned to when it exits Rx; `PKT_DONE_EXIT_EN` is used to determine when receiving `PKT_DONE` whether it will exit Rx and return to the state configured by `RX_EXIT_STATE` or keep in the current state.

After having configurations done, users can set `RX_AUTO_HOP_EN` to 1 to enable the function, then send command `go_rx` to enter the `RX_Auto_Hop` mode. During the auto-operation, if events triggering the stop of the auto-operation occur, the chip will stop the auto-operation and stay in some fixed state. If requiring to stop the `RX_Auto_Hop` operation, users can set `API_STOP` to 1. When the chip detects this bit being set, it will set `API_DONE_FLAG` to 1 upon the next state switching, then keep the current configurations, exit the `RX_Auto_Hop` function and return to READY state.

After the halt of `RX Auto Hop`, if `RX_AUTO_HOP_EN` is kept as 1, the chip will enter Rx-auto-hopping operation again once users send `go_rx` command again, and it will start hopping from the next frequency point since the latest finished hopped frequency point. If users wanted start hopping from the first frequency point, they can clear register `FREQ_DONE_TIMES` before sending `go_rx` command.

The Rx-Auto-Hop supports the following 7 operating modes.

Table 18. Rx-Auto-Hop Operating Mode

No.	Rx Extension Mode	Rx Extension Condition
0	If it is set to 0, there's no Rx extension. If PKT_DONE is detected before T1 is timeout and PKT_DONE_EXIT_EN = 1, it exits auto-hopping and switches to the state specified by RX_EXIT_STATE. Otherwise, it exits Rx when T1 is timeout and returns to the state specified by FREQ_SW_STATE then continue to hop the next frequency point.	None
1	If it meets detection conditions within T1, it exits from T1 and hands over control to the MCU (and keeps the state); if it does not meet detection conditions within T1, it exits Rx when T1 is timeout and returns to the state specified by register FREQ_SW_STATE and continue to hop the next frequency point.	RSSI_PJD_VLD is valid.
2		PREAM_OK is valid.
3		RSSI_PJD_VLD and PREAM_OK are valid at the same time.
4		RSSI_PJD_VLD is valid.
5	If it meets detection conditions within T1, it switches to T2; otherwise if it does not meet detection conditions and timeout occurs, then it exits Rx and returns to the next frequency point. If it detects SYNC within T2, it exits from T2 and hands over control to the MCU (and keeps the state); otherwise it exits Rx when T2 is timeout and returns to the state specified by register FREQ_SW_STATE and continue to hop to the next frequency point.	PREAM_OK is valid.
6		RSSI_PJD_VLD and PREAM_OK are valid at the same time.

7.4 TX Automatic Frequency Hopping (TX AUTO HOP)

In TX-Auto-Hop, FREQ_TIMES<7:0> is used to configure the size of the frequency hopping table. The table contains a maximum of 64 bytes of space for users to store the channels that need to be configured. The frequency hopping process is as follows.

The chip enters a channel for transmission. After a successful transmission, the chip returns to the READY or TFS state according to FREQ_SW_STATE, automatically configures the channel value of the next byte in the frequency hopping table, and then re-enters the TX state for transmission. If the automatic switching state is READY, the PLL frequency will be calibrated before re-entering TX, which will consume the corresponding calibration time. If the automatic switching state is TFS, the PLL frequency will not be calibrated, and the switching time will be shorter. If the interval between any two frequency points in the frequency hopping table exceeds 2MHz, it is recommended to configure the FREQ_SW_STATE to the READY state because it is necessary to re-calibrate the PLL.

When it finishes the entire table hopping, if RX_HOP_PERSIST is configured to 1, the chip will continue to hop the table from the beginning; otherwise, if RX_HOP_PERSIST is configured to 0, it will return to the TX_EXIT_STATE state and stop the operation of automatic frequency hopping transmission. The formula to calculate the target frequency in frequency hopping is:

$$\text{FREQ} = 1K \times \text{FREQ_SPACE}<7:0> \times \text{FREQ_CHANL}<7:0>$$

Where, FREQ_SPACE<7:0> is an independent register, and FREQ_CHANL<7:0> is configured by 64 registers, each storing one

channel value.

The TX_Auto_Hop mode provides the REQ_CHANL_LV register to inform users of the currently used channel value. Additionally, there is the FREQ_DONE_TIMES register, which is used to read the number of frequency hopping times that have been completed.

When making the configurations ready, users can set TX_AUTO_HOP_EN to 1 to enable the function, then send the go_tx command to enter the TX_Auto_Hop mode. If requiring to stop the TX_Auto_Hop function forcibly, users can set API_STOP to 1; then the chip detects this bit and subsequently upon the next state switching it sets API_DONE_FLAG to 1 with retaining the current configurations, thereby, it exits TX_Auto_Hop and returns to the READY state.

7.5 Automatically Re-sending (TX AUTO RESEND)

The automatic re-sending function is an automatic control mechanism to realize the handshake between the transmitter and the receiver. It provides two packet format fields, SEQNUM and FCS2, to judge if the handshaking between transmitter and the receiver succeeds. These two fields are part of the packet, and need to be enabled by SEQNUM_EN and FCS2_EN when the function is used.

Its implementation principle is that: the transmitter configures the local SEQNUM and sets the NACK bit in the FCS2 field to 1 for transmission; after the transmission is accomplished, it enters the RX state and waits for the response; when it receives the response and finds the SEQNUM in the response packet matches the local SEQNUM and the received NACK bit is 1, the handshaking is recognized successful.

At the receiver side, after enabling the ACK function, it enters the RX state. When the packet is received successfully, the receiver judges whether need to answer it by the received NACK bit. If a response is needed, it enters the TX state, and the received sequence number and NACK bit are sent back. The PKT_DONE interrupt is generated after sending back operation is completed; then it either returns back to the state configured in RX_EXIT_STATE or remains in the RX state according to PKT_DONE_EXIT_EN.

7.6 Carrier Sensing Multi-path Accessing (CSMA)

CSMA is a kind of conflict avoidance mechanism (speaking after sensing), that is, before transmission it senses the channel idle status first, then determine whether to transmit or not, which helps to avoid the potential conflicts when multiple transmitters use the same channel and increase the possibility of successful packet receiving for receivers. The key point of CSMA is comparing the RSSI with the specified threshold to judge the channel idle status. When the channel is busy, it will repeat the procedure from back-off to listening for several times until it finds the channel in idle. If it fails to detect channel being idle when reaching the repeated time limit, the corresponding interrupt and flag will be generated and transmission will be stopped. During back-off procedure, the chip stays in SLEEP state. In the PERSISTENT mode, it will back to RFS state first and then switch to Rx state for listening rather than back to SLEEP state. If it was under non-PERSISTENT mode, the maximum number of non-idle channels can be configured by the CSMA_TIMES. While as if the channel was still busy, it will exit the CSMA and return to the READY state.

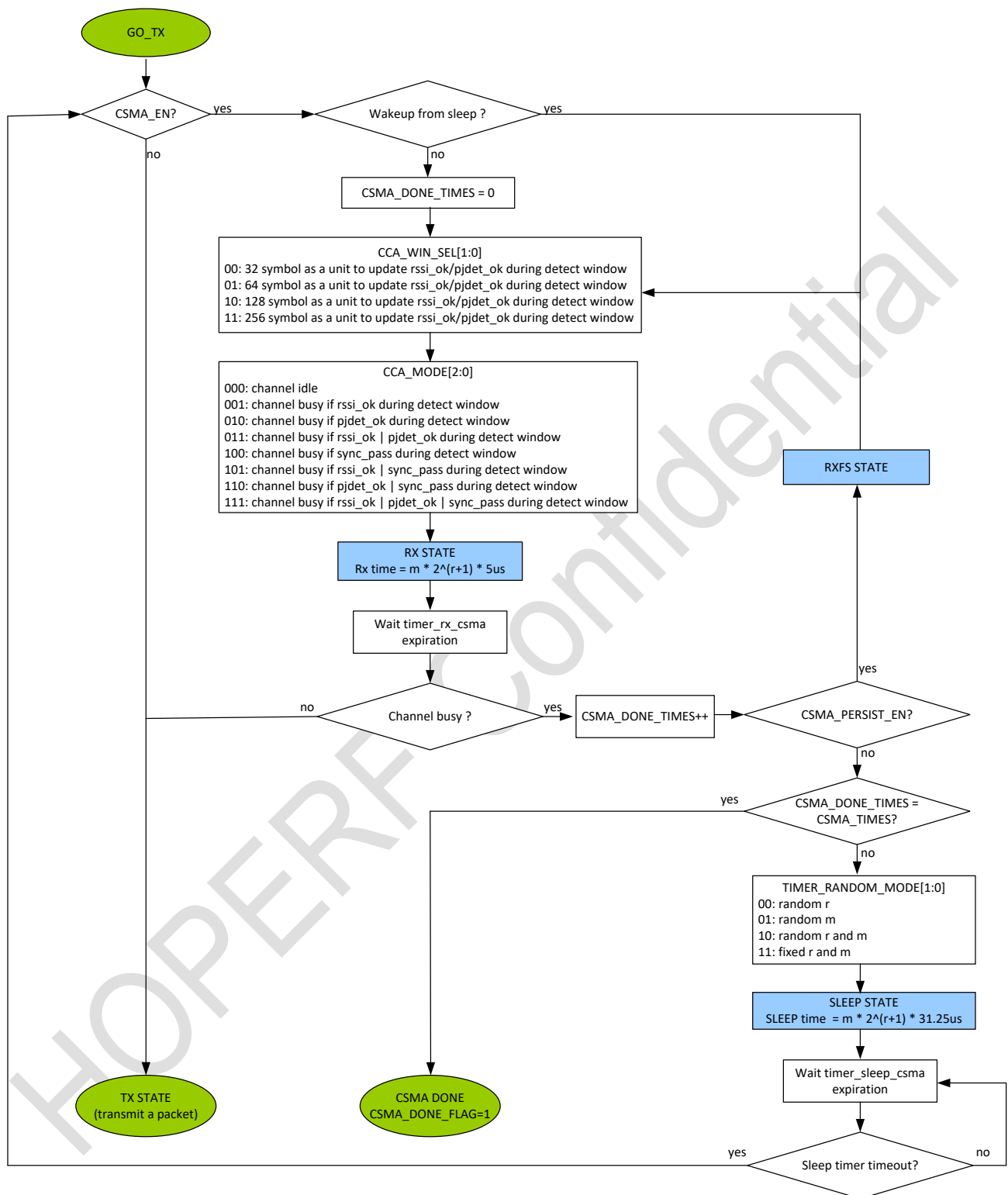


Figure 30. Operation Process of Carrier Sensing Multi-path Accessing

7.7 Antenna Diversity

The mutual influence of direct waves, reflected waves, and scattered waves generated by various objects in the transmission path of radio frequency signals (i.e., multipath fading), as well as the loss caused by Doppler frequency shift (the wavelength of the radiation from an object changes due to the relative motion between the wave source and the observer), can be improved through diversity techniques. Diversity techniques include spatial diversity, polarization diversity, angle diversity, frequency diversity, and time diversity. The CMT2312A provides dual - antenna angle diversity technology, enabling electromagnetic waves to reach the receiving end through two different paths at different angles. The receiving end of the CMT2312A uses RSSI to determine the signal - to - noise ratio of the two paths and then decides which path to use for packet reception.

Users can configure the ANT_DIV_MANU register to 0 to let the CMT2312A automatically switch and lock the antenna. Configure ANT_DIV_EN to 1 to enable the antenna diversity function. When ANT_SW_DIS is configured to 0, when the two - antenna switching scan finds that the locking conditions are met, the RSSI values of the two antennas are compared, and the antenna with the larger RSSI value is automatically locked. When ANT_SW_DIS is configured to 1, when the two - antenna switching scan finds that the current RSSI value is 16 dB larger than the RSSI_Compare_TH (the register is RSSI_ABSOLUTE_TH) set on the RFPDK, the current antenna is locked and no longer switched; otherwise, the RSSI values of the two antennas are compared before locking.

Users can also set the ANT_DIV_MANU register to 1 to manually control the antenna diversity switch. When ANT_SELECT is set to 0, GPIO2 outputs 1 and GPIO3 outputs 0. When ANT_SELECT is set to 1, GPIO2 outputs 0 and GPIO3 outputs 1.

The application schematic diagram of the antenna diversity transceiver using the recommended direct connection matching network is shown in the following figure. The two antennas are placed at a right angle, and the antenna switch is automatically controlled by GPIO2 and GPIO3. When GPIO2 outputs 1, GPIO3 outputs 0, and the antenna controlled by GPIO2 is selected. Conversely, when GPIO2 outputs 0, GPIO3 outputs 1, and the antenna controlled by GPIO3 is selected. An antenna switch deteriorates the noise figure of the receiver by approximately 0.5dB.

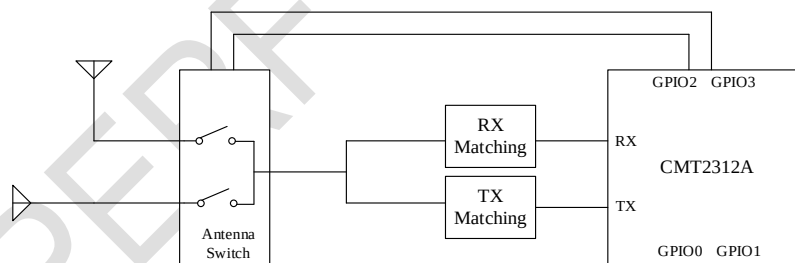


Figure 31. Application Schematic Diagram of an Antenna Diversity Transceiver Using a Direct Connection Matching Network.

If the customer needs to design the receiving and transmitting matching networks separately, as shown in the figure below, add an antenna switch, which is controlled by GPIO0 and GPIO1, and configure TRX_SWITCH_EN = 1 to enable the TRX switch. When TRX_SWITCH_INV = 0, in the RX state, GPIO0 outputs 1 and GPIO1 outputs 0. In the TX state, GPIO0 outputs 0 and GPIO1 outputs 1. If TRX_SWITCH_INV = 1, the above output values are reversed.

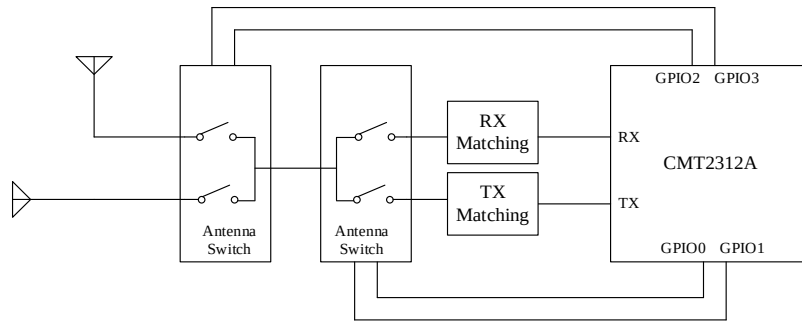


Figure 32. Application Schematic Diagram of an Antenna Diversity Transceiver using a Separate Matching Network.

ANT_LOCK_EN is the enabling bit of the diversity antenna lock interrupt. ANT_LOCK_FLAG is the antenna lock flag bit. ANT_LOCK_CLR is the antenna lock interrupt flag clear bit. Users can read the value on the register ANT_INSTR to know which antenna is automatically locked to the current antenna diversity when the antenna locked. If ANT_INSTR is 0, it will lock the GPIO2= 1&GPIO3 =0 connected antenna; otherwise, it will be GPIO2= 0&GPIO3 =1.

Users can also read the RSSI values of the two antennas after the lock by using the RSSI_VALUE and RSSI_MIN_VALUE registers. RSSI_VALUE is the large value while RSSI_MIN_VALUE is the small one. It is recommended that users select sync_ok for RSSI Detect Mode in the RFPDK Feature 2 Settings page, so that the locked antenna will update the RSSI value to the RSSI_VALUE after sync_ok.

7.8 Quick Switch Pre-saved Configuration

The CMT2312A supports the pre - storage of up to 7 sets of configurations. By operating the API_CMD instruction, any specified set among the 7 sets of configurations can be switched. This is a batch read - write at the chip - internal DMA level, which takes less than 1:50uS. Compared with the external master controller updating all configurations again through SPI, this internal switching method is much faster. It is suitable for application scenarios where different configurations need to be switched frequently and there are requirements for switching time. For the specific usage, please refer to the related document AN244 "CMT2312A Quick Switching of Pre - stored Configuration Usage Guide".

8 User Register

All configuration and control operations from external MCU are accomplished by accessing the registers in the 3 pages through SPI interface. Shifting among the 3 pages will be accomplished through writing the address 0x7E. As shown in the table below, the address range 0x00 ~ 0x71 are divided into 3 banks for understanding purpose, configuration bank (including 6 sub-banks), control bank 1 and control bank 2, which will be detailed below.

The addresses of the 3 banks are continuous. Operation method against the 3 banks have no essential difference, namely all operations are direct read/write operations following the SPI register read/write timing. However, for the 3 banks the functionalities and usage are different as shown in the table below.

Table 19. CMT2312A Register Bank Partitioning

Page	Address	Bank	Function Description
0	0x00 - 0x0F	System control bank 1	For chip operation and control like state switching, status checking, etc.
	0x10 - 0x17	System control bank 2	For chip function enabling.
	0x18 - 0x27	Interrupt Control bank	For reading and clearing interrupt flag, inaccessible in SLEEP state.
	0x28 - 0x5F	Configuration bank (packet format)	For packet format and FIFO related configurations.
	0x60 - 0x77	Configuration bank (system feature)	For system operating related function configurations.
1	0x00 - 0x0F	Configuration bank (CMT internal use)	For CMT internal use only.
	0x10 - 0x27	Configuration bank (Tx)	For transmitter related feature configurations.
	0x30 - 0x68	Configuration bank (Rx)	For receiver related feature configurations.
2	0x00 - 0x3F	Configuration bank (auto-frequency hopping)	For storing automatic channel hopping table information.
Notes: 1. In Page 0, address 0x7A is FIFO accessing interface address, address 0x7B is the interface for continuously accessing register. 2. In all Pages, address 0x7E is the interface for Page shifting. 3. In all Pages, address 0x7F is the interface for software reset. 4. The addresses unlisted in the table are inaccessible.			

Except the interrupt configuration bank, all banks can be accessed in SLEEP state and the configuration data will retain if no POR reset or power down occurs.

The register values in Page 0 and Page 1 can come either from RFPDK or from user application based on actual requirements. In general, except a few parameters about RF frequency or data rate that may require configuring for multiple times in user application, most of the registers just need to configure once in the initial stage.

As for the frequency hopping table in Page 2, users need to have specific design and configuration according to the guidance in related AN document.

9 Ordering Information

Table 20. CMT2312A Ordering Information

Model	Description	Packaging	Package	Operating Condition	Minimum Order Quantity
CMT2312A-EQR ^[1]	CMT2312A, ultra-low power sub-1GHz RF transceiver	QFN24 (4x4)	Tape and Reel	1.8 to 3.6V, -40 to 85 °C	3,000 pcs
Notes: [1]. E refers to extended Industrial product rating, which supports temperature range from -40 to +85 °C. Q refers to the package type QFN24 4x4. R refers to tape and reel type, and the minimum ordering quantity (MOQ) is 3,000 pieces.					

Please visit www.hoperf.com for more product/product line information.

Please contact sales@hoperf.com or your local sales representative for sales or pricing requirements.

10 Packaging Information

The packaging information of the CMT2312A is shown in the figure below.

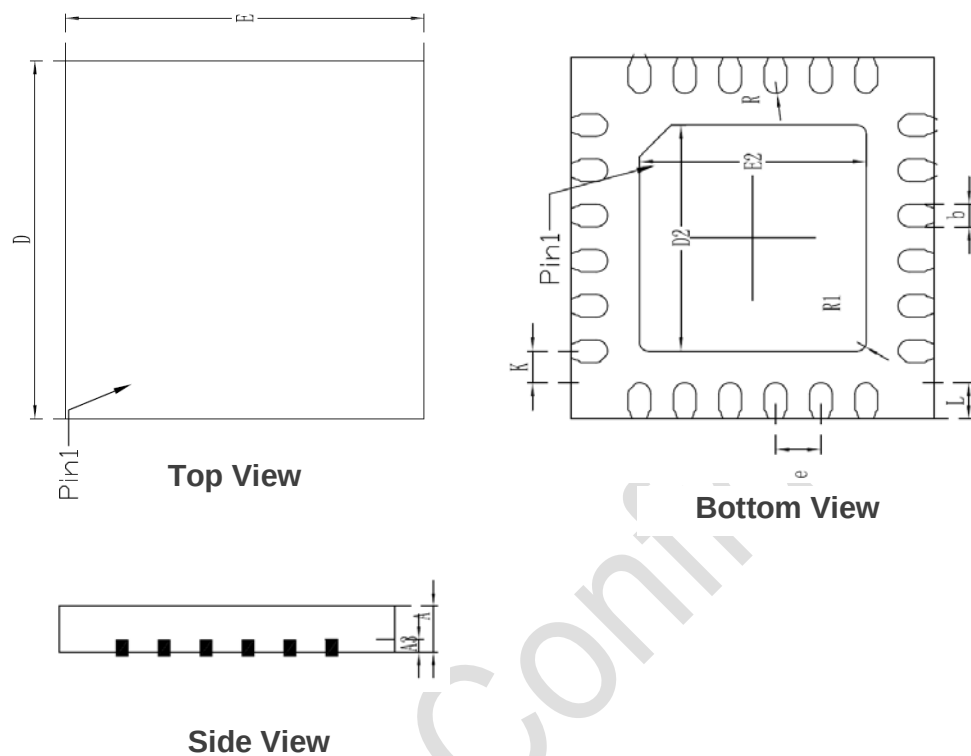


Figure 33. 24-Pin QFN 4x4 Packaging

Table 21. 24-Pin QFN 4x4 Packaging Scale

Symbol	Scale (mm)		
	Min.	Typ.	Max.
A	0.70	0.75	0.80
A1	0.00	0,02	0,05
A3	--	0.15	--
b	0.20	0.25	0.30
D	3.90	4.00	4.10
E	3.90	4.00	4.10
e	0.40	0.50	0.60
D2	2.60	2.70	2.80
E2	2.60	2.70	2.80
L	0.20	0.30	0.40
K	0.20	--	-
R	--	0.10	-
R1	-	0.10	-

11 Top Marking

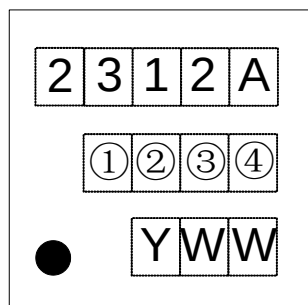


Figure 34. CMT2312A Top Marking

Table 22. CMT2312A Top Marking Information

Marking method	Laser.
Pin 1 mark	Diameter of the circle = 0.3 mm.
Fontsize	0.5 mm, align right.
Line 1 marking	2312A, refers to model CMT2312A.
Line 2 marking	①②③④ is internal tracing code.
Line 3 marking	YWW is the date code is assigned by assembly factory. Y represents the last digit of the year. WW represents working week.

12 Revise History

Table 23. Revise History Records

Version No.	Chapter	Description	Date
1.0	All	Initial version	2025-08-12

13 Contacts

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